INN HOTELS – CASE STUDY

Project Report

Project Details:

Project Name	INN Hotels - Case study
Original Author	Reshma B
Date	11-08-2024

greatlearning Learning for Life

List of Tables

List of Figures

greatlearning
Learning for Life

Contents

greatlearning Learning for Life

greatlearning
Learning for Life

greatlearning
Learning for Life

1. Project Background

A significant number of hotel bookings are called off due to cancellations or no-shows. The typical reasons for cancellations include change of plans, scheduling conflicts, etc. This is often made easier by the option to do so free of charge or preferably at a low cost which is beneficial to hotel guests but it is a less desirable and possibly revenue-diminishing factor for hotels to deal with. Such losses are particularly high on last-minute cancellations. The new technologies involving online booking channels have dramatically changed customers' booking possibilities and behaviour. This adds a further dimension to the challenge of how hotels handle cancellations, which are no longer limited to traditional booking and guest characteristics.

The cancellation of bookings impacts a hotel on various fronts:

1. Loss of resources (revenue) when the hotel cannot resell the room.
2. Additional costs of distribution channels by increasing commissions or paying for publicity to help sell these rooms.
3. Lowering prices last minute, so the hotel can resell a room, resulting in reducing the profit margin.
4. Human resources to make arrangements for the guests.

2. Objective

INN Hotels Group has a chain of hotels in Portugal, they are facing problems with the high number of booking cancellations. The increasing number of cancellations calls for a Machine Learning based solution that can help in predicting which booking is likely to be cancelled.

The primary focus will be on:

- Analysing the data provided to find which factors have a high influence on booking cancellations.
- Building a predictive model that can predict which booking is going to be cancelled in advance.
- Helping in formulating profitable policies for cancellations and refunds.

3. Data Description

The data contains the different attributes of customers' booking details. The detailed data dictionary is given below.

- Booking_ID: the unique identifier of each booking
- no_of_adults: Number of adults
- no_of_children: Number of Children
- no_of_weekend_nights: Number of weekend nights (Saturday or Sunday) the guest stayed or booked to stay at the hotel
- no_of_week_nights: Number of weeknights (Monday to Friday) the guest stayed or booked to stay at the hotel
- type_of_meal_plan: Type of meal plan booked by the customer:
  Not Selected – No meal plan selected
  Meal Plan 1 – Breakfast
  Meal Plan 2 – Half board (breakfast and one other meal)
  Meal Plan 3 – Full board (breakfast, lunch, and dinner)
- required_car_parking_space: Does the customer require a car parking space? (0 - No, 1- Yes)
- room_type_reserved: Type of room reserved by the customer. The values are ciphered (encoded) by INN Hotels Group
- lead_time: Number of days between the date of booking and the arrival date
- arrival_year: Year of arrival date
- arrival_month: Month of arrival date
- arrival_date: Date of the month
- market_segment_type: Market segment designation.
- repeated_guest: Is the customer a repeated guest? (0 - No, 1- Yes)
- no_of_previous_cancellations: Number of previous bookings that were cancelled by the customer prior to the current booking
- no_of_previous_bookings_not_canceled: Number of previous bookings not cancelled by the customer prior to the current booking
- avg_price_per_room: Average price per day of the reservation; prices of the rooms are dynamic. (in euros)
- no_of_special_requests: Total number of special requests made by the customer (e.g. high floor, view from the room, etc)
- booking_status: Flag indicating if the booking was cancelled or not.

4. Exploratory Data Analysis

4.1 Data structure

After importing NumPy and Pandas libraries for tasks like data reading, data computation and data manipulation and loading data, the following observations regarding the data structure were noted.

	Booking_ID	no_of_adults	no_of_children	no_of_weekend_nights	no_of_week_nights	type_of_meal_plan	required_car_parking_space	room_type_reserved	lead_time
0	INN00001	2	0	1	2	Meal Plan 1	0	Room_Type 1	224
1	INN00002	2	0	2	3	Not Selected	0	Room_Type 1	5
2	INN00003	1	0	2	1	Meal Plan 1	0	Room_Type 1	1
3	INN00004	2	0	0	2	Meal Plan 1	0	Room_Type 1	211
4	INN00005	2	0	1	1	Not Selected	0	Room_Type 1	48

arrival_year	arrival_month	arrival_date	market_segment_type	repeated_guest	no_of_previous_cancellations	no_of_previous_bookings_not_canceled	avg_price_per_room
2017	10	2	Offline	0	0	0	65.00
2018	11	6	Online	0	0	0	106.68
2018	2	28	Online	0	0	0	60.00
2018	5	20	Online	0	0	0	100.00
2018	4	11	Online	0	0	0	94.50

no_of_special_requests	booking_status
0	Not_Canceled
1	Not_Canceled
0	Canceled
0	Canceled
0	Canceled

OBSERVATION	VARIABLES
36275	19

4.2 Data types

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 36275 entries, 0 to 36274
Data columns (total 19 columns):
 #   Column                                Non-Null Count  Dtype
---  ------                                --------------  -----
 0   Booking_ID                            36275 non-null  object
 1   no_of_adults                          36275 non-null  int64
 2   no_of_children                        36275 non-null  int64
 3   no_of_weekend_nights                  36275 non-null  int64
 4   no_of_week_nights                     36275 non-null  int64
 5   type_of_meal_plan                     36275 non-null  object
 6   required_car_parking_space            36275 non-null  int64
 7   room_type_reserved                    36275 non-null  object
 8   lead_time                             36275 non-null  int64
 9   arrival_year                          36275 non-null  int64
 10  arrival_month                         36275 non-null  int64
 11  arrival_date                          36275 non-null  int64
 12  market_segment_type                   36275 non-null  object
 13  repeated_guest                        36275 non-null  int64
 14  no_of_previous_cancellations          36275 non-null  int64
 15  no_of_previous_bookings_not_canceled  36275 non-null  int64
 16  avg_price_per_room                    36275 non-null  float64
 17  no_of_special_requests                36275 non-null  int64
 18  booking_status                        36275 non-null  object
dtypes: float64(1), int64(13), object(5)
memory usage: 5.3+ MB
```

Table 1. Data types

- The data set contains nineteen variables indexed from 0-18.
- The data set consists of 14 numerical columns, where 13 are of integer data type and 1 (avg_price_per_room) is float data type and 5 object columns.
- The below columns can be treated as categorical columns:
    1. A column named 'required_car_parking_space' which is in int data type given in terms of 0 and 1, which indicates if a customer does not require or require a parking space respectively.
    2. A column named 'repeated_guest' which is in int data type given in terms of 0 and 1, which indicates if the customer is not a repeated guest or is a repeated guest respectively.
    3. A column named arrival_year which gives the year of arrival of the customer.

4. A column named arrival_month which gives the month of arrival of the customer.
5. A column named arrival_date which gives the date of arrival of the customer.

4.3 Null values

- No null values were found in the data set.

4.4 Duplicate Values

- No duplicate values were found in the data set.

4.5 Statistical Summary

The following observations were made:

- no_of_adults: The number of adults range from 2-4, which is acceptable.
- no_of_children: The number of children ranges from 0-10, which implies the presence of outliers.
- no_of_weekend_nights: On average, guests are expected to stay for 1 weekend night, but they may stay for up to 7 weekend nights, indicating the presence of outliers.
- no_of_week_nights: The number of weekday nights reaches up to 3 at the 75th percentile, but there are instances of up to 17 weekday nights, indicating the presence of outliers.
- type_of_meal_plan: Out of the 4 meal plans, Meal plan 1 seems to the top with a frequency of 27835.
- room_type_reserved: Among the 7 room types, Room Type 1 leads with a frequency of 28130.
- lead_time: On average, the lead time is 85 days, with a maximum of 443 days.
- market_segment_type: The primary market segment is online, with a count of 23214.
- no_of_previous_cancellations: The number of previous booking cancellation ranges from 0-13 and the number of previous bookings not cancellation ranges from 0-58.
- no_of_previous_bookings_not_canceled: Approximately 50% of the rooms cost 100 euros on an average.
- avg_price_per_room: The number of special requests made by customers ranges from 0 to 5.
- booking_status: Out of 36275, the booking was not cancelled 24390 times.

4.6 Count and Percentage of Categorical Variable in the Data Set

The highest and lowest scoring variables of the categorical variable are given in the data below:

TYPE OF MEAL PLAN	COUNT	PERCENTAGE
MEAL PLAN 1	27835	76
MEAL PLAN 3	5	0.013

CAR PARKING SPACE	COUNT	PERCENTAGE
NO	35151	96
YES	1124	3

ROOM TYPE	COUNT	PERCENTAGE
ROOM TYPE 1	28130	77
ROOM TYPE 3	7	0.019

MARKET SEGMENT	COUNT	PERCENTAGE
ONLINE	23214	63
AVIATION	125	0.34

REPEATED GUEST	COUNT	PERCENTAGE
NO	35345	97
YES	930	2.5

BOOKING STATUS	COUNT	PERCENTAGE
NOT CANCELED	24390	67
CANCELED	6514	32

ARRIVAL YEAR	COUNT	PERCENTAGE
2018	29761	82
2017	6514	17

ARRIVAL MONTH	COUNT	PERCENTAGE
10	5317	14
1	1014	2.7

ARRIVAL DATE	COUNT	PERCENTAGE
13	1358	3.7
31	578	1.5

Table 2. Count and percentage of categorical variables

4.7 Univariate analysis

4.7.1 Number of adults

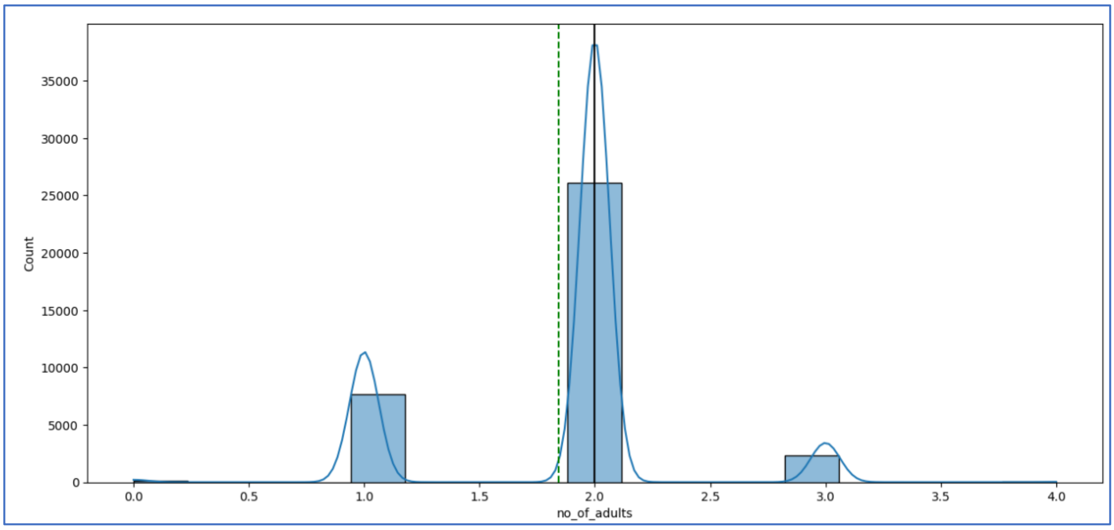


Fig 1. Number of adults

OBSERVATIONS:

- The mean and median are almost close to each other, suggesting a normal distribution.
- The highest number of bookings is for 2 adults, followed by 1 adult and 3 adults.

4.7.2 Number of children

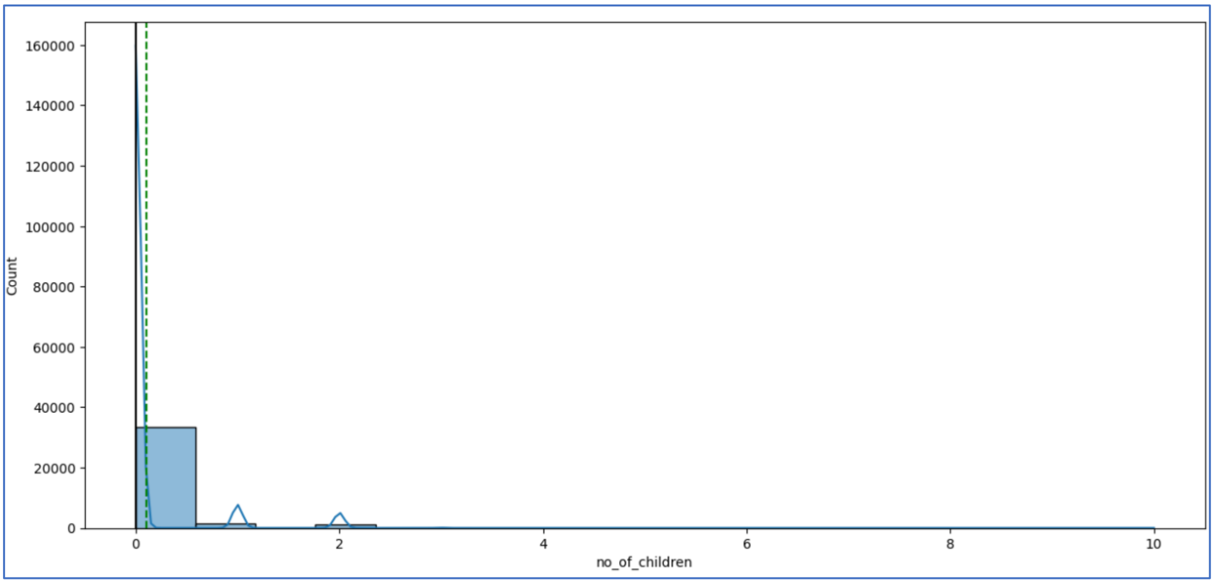


Fig 2. Number of children

OBSERVATIONS:

- The number of children is a highly right skewed distribution.
- The majority of bookings do not include children.
- However, the number of children range from 0-10.

4.7.3 Number of weekend nights

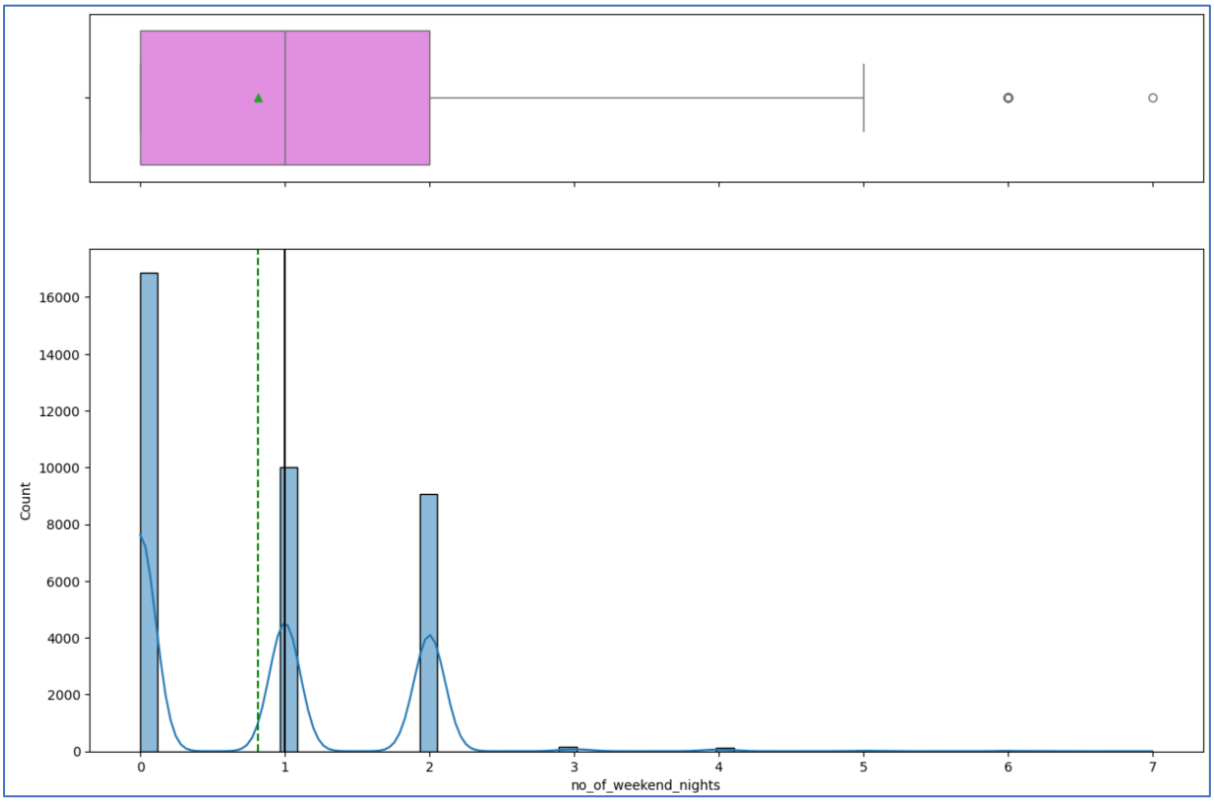


Fig 3. Number of weekend nights

OBSERVATIONS:

- The number of weekend nights is a highly right skewed distribution.
- Most bookings do not include weekend nights, but those with 1-2 weekend nights are more frequent.
- However, the number of weekend nights range from 0-7.

4.7.4 Number of week nights

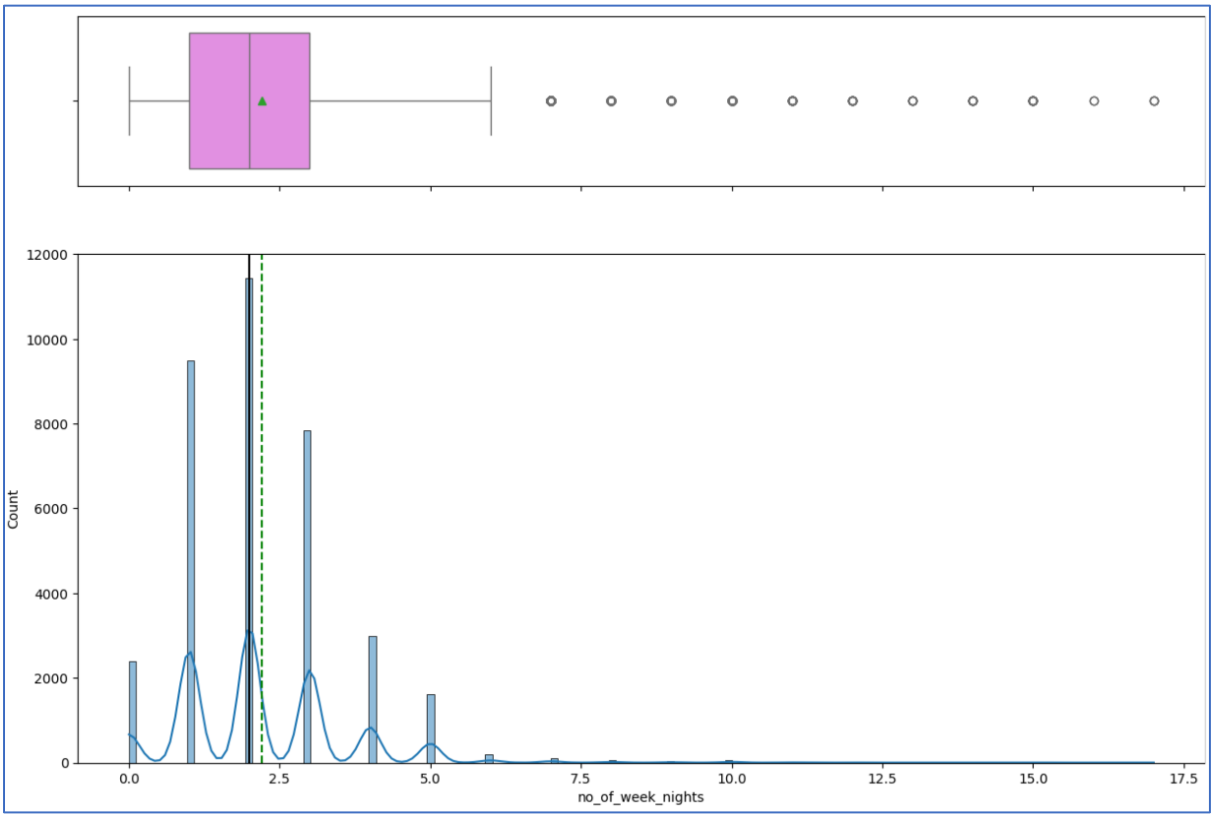


Fig 4. Number of week nights

OBSERVATIONS:

- The mean and median are close to each other, indicating a normal distribution.
- Most bookings include 2-week nights, with 1 and 3-week nights being the next most common.
- Bookings with 4-5-week nights are also quite common.
- Bookings extending beyond 7 days, up to 17 days, are also observed.

4.7.5 Lead time

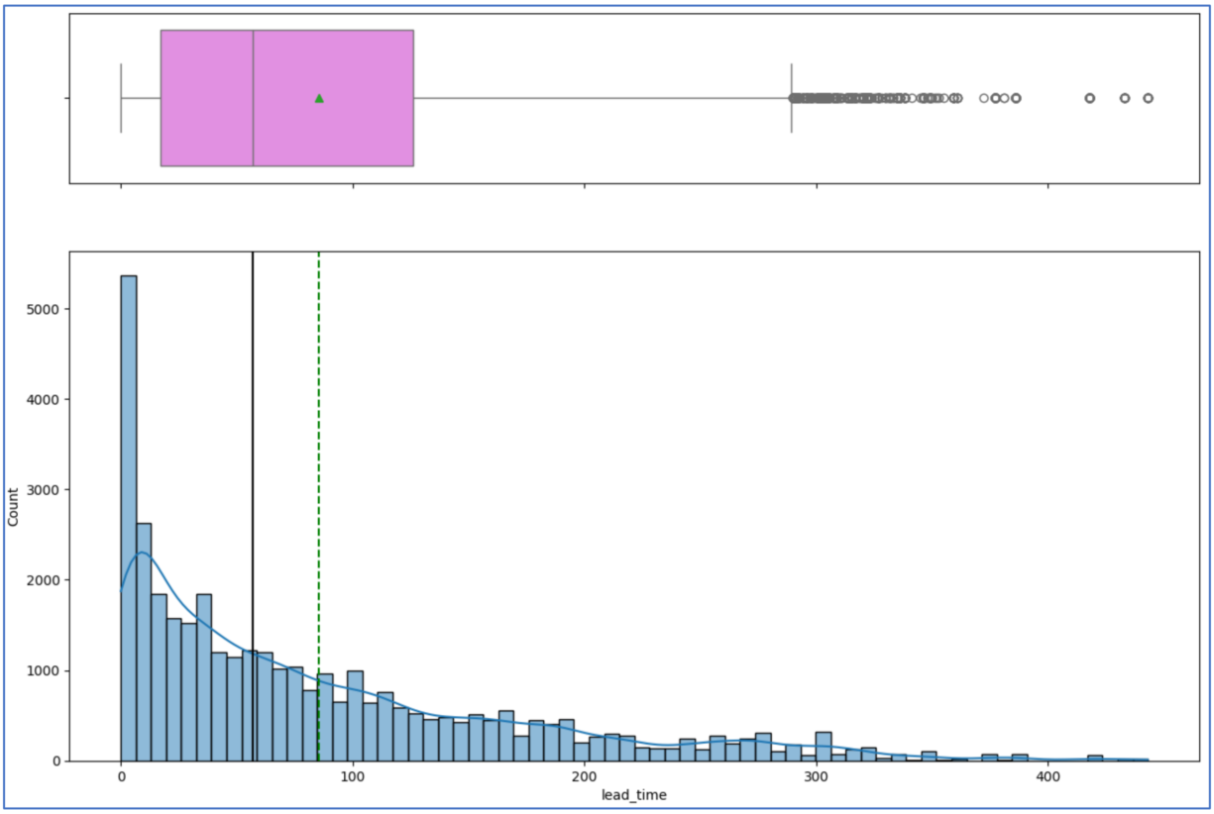


Fig 5. Lead time

OBSERVATIONS:

- The lead time is a highly right skewed distribution.
- The lead time ranges from 0 to 430 days, with an average of 90 days.
- The outliers fall between 300 and 430 days.

4.7.6 Number of previous bookings cancelled

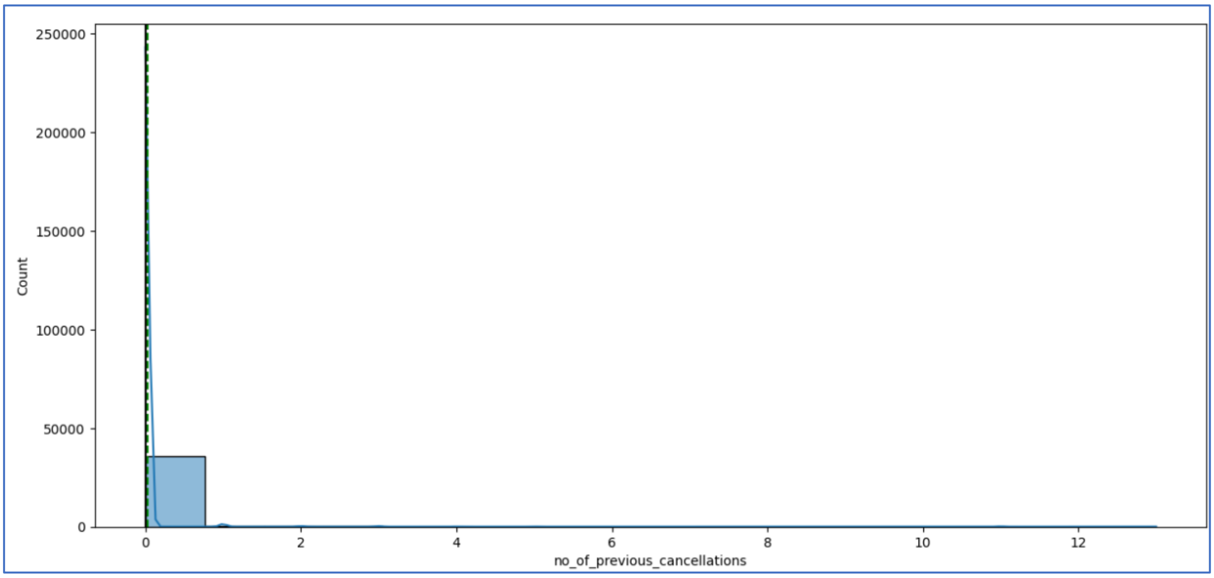


Fig 6. Number of previous bookings cancelled

OBSERVATIONS:

- The number of previous bookings that were cancelled by the customer prior to the current booking ranges from 0-13.
- The number of bookings that are not cancelled prior to the current booking is higher.

4.7.7 Number of previous bookings not cancelled

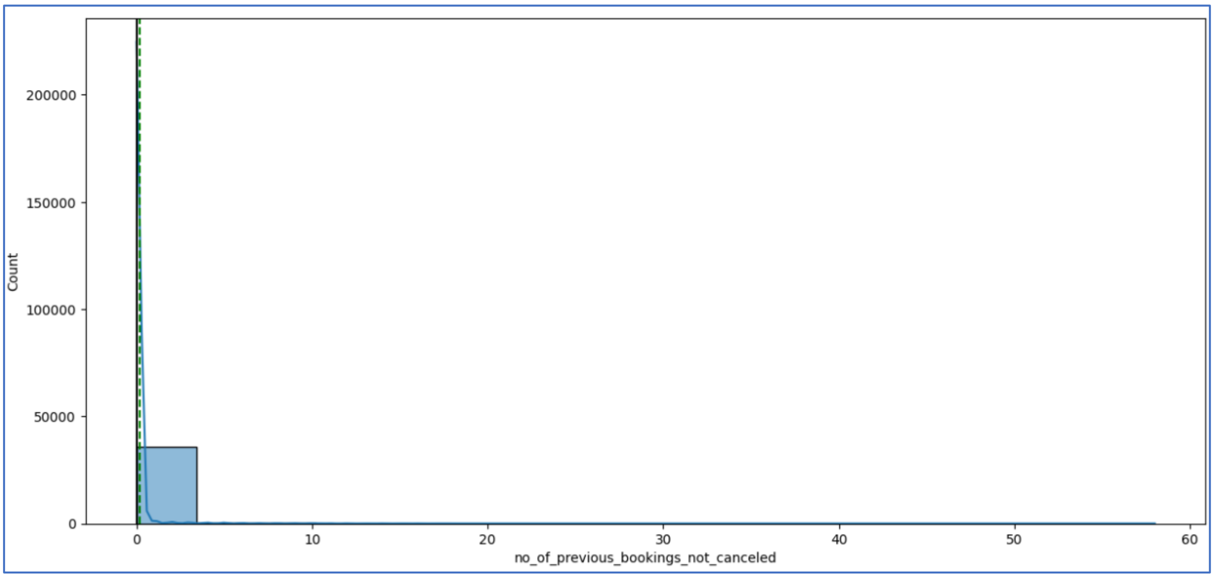


Fig 7. Number of previous bookings not cancelled

OBSERVATIONS:

- The number of previous bookings that were not cancelled by the customer before the current booking ranges from 0 to 58.
- The number of bookings cancelled before the current booking is higher than the number of those that were not cancelled previously.

4.7.8 Average price per room

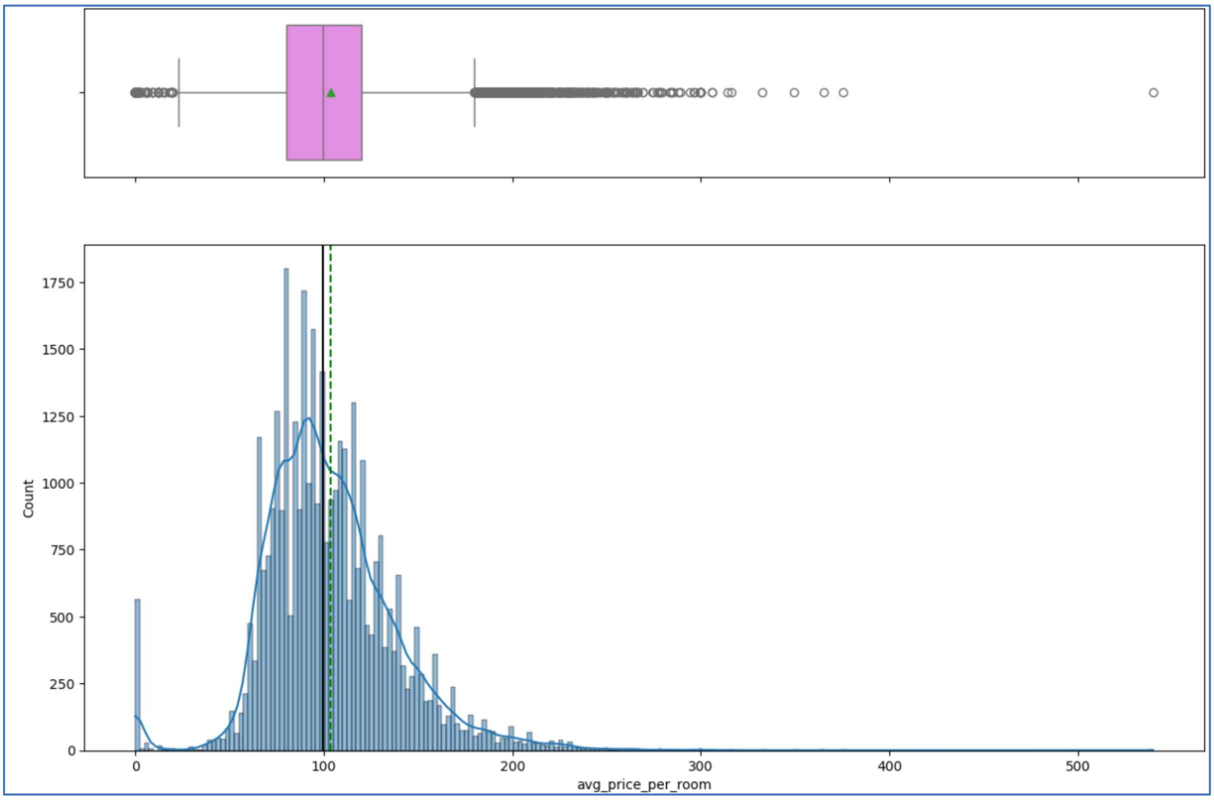


Fig 8. Average price per room

OBSERVATIONS:

- The average price per day of the reservation shows a normal distribution.
- The mean and median are close to each other.
- The outliers range from 180-520 euros.
- The average price per day of the reservation is 100 euros.

4.7.9 Number of special requests

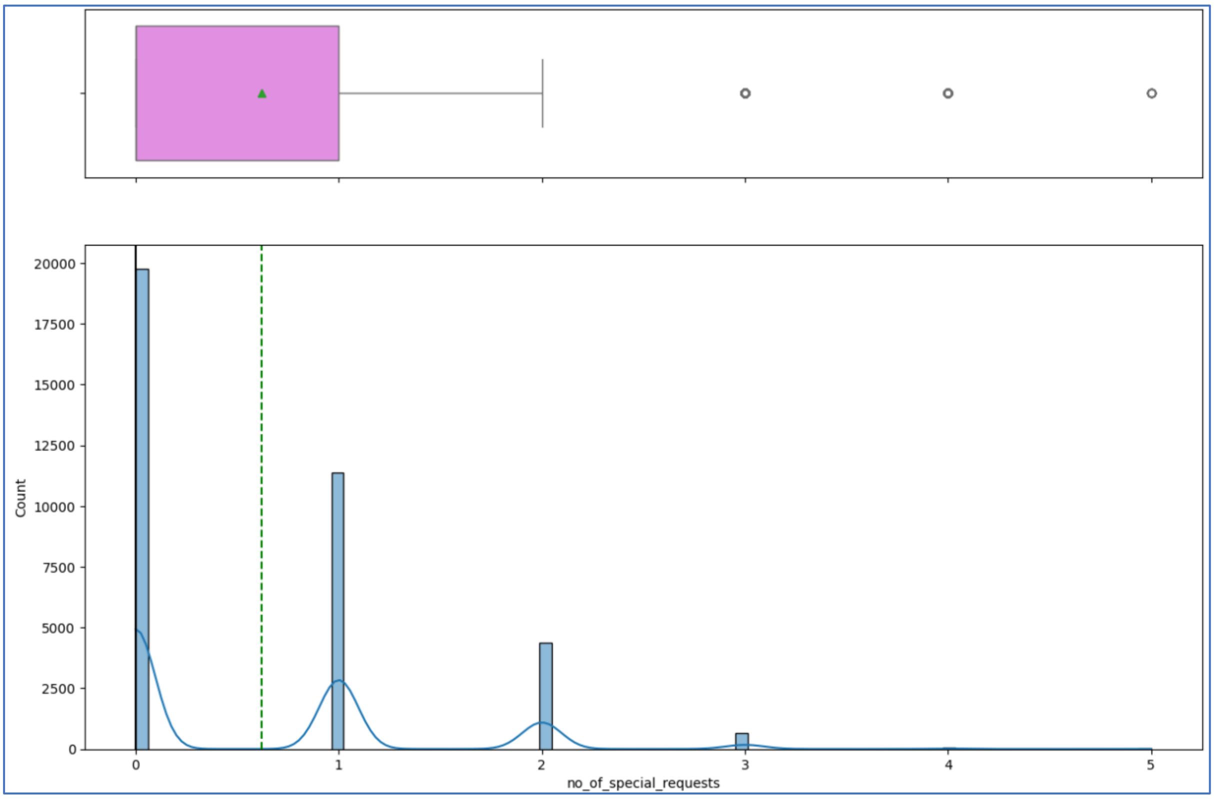


Fig 9. Number of special requests

OBSERVATIONS:

- The total number of special requests raised by the customer range from 0-5.
- Most customers have made only one request, followed by those who have made 1, 2, or 3 requests.

4.7.10 Type of meal plan

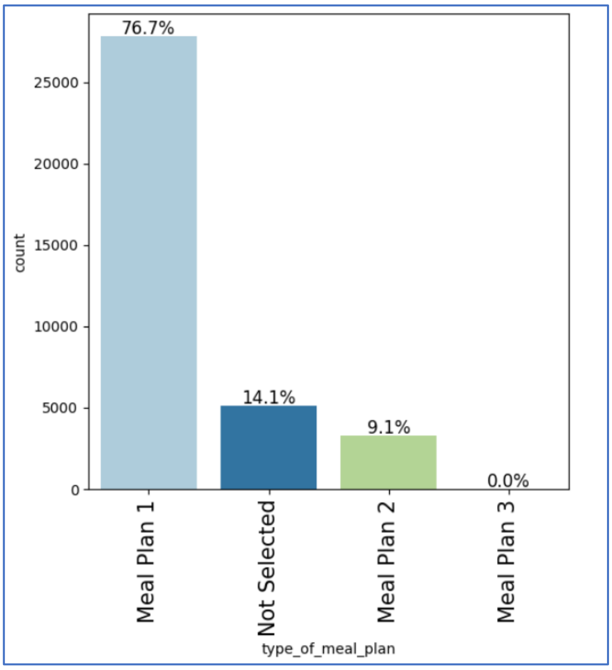


Fig 10. Type of meal plan

OBSERVATIONS:

- Out of the 3 meal plans, majority of customers (76.7%) have selected meal plan 1.
- The second highest proportion of customers, accounting for 14.1%, did not choose the meal plan.
- Meal Plan 3 is the least selected option.

4.7.11 Required car parking space

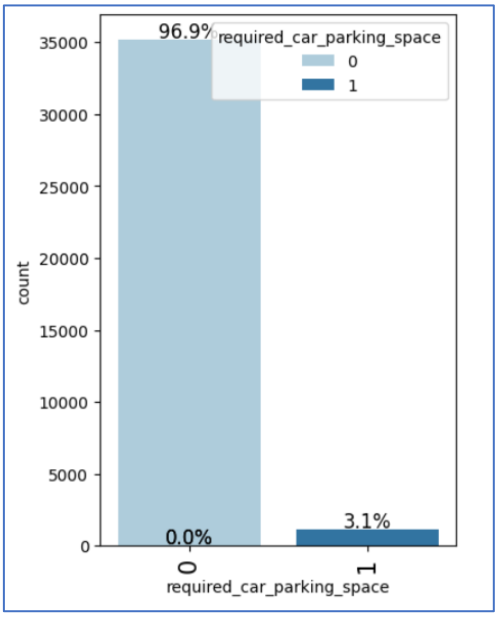


Fig 11. Required car parking space

OBSERVATIONS:

- The majority of customers, 96.9%, require a car parking space.

4.7.12 Room type reserved

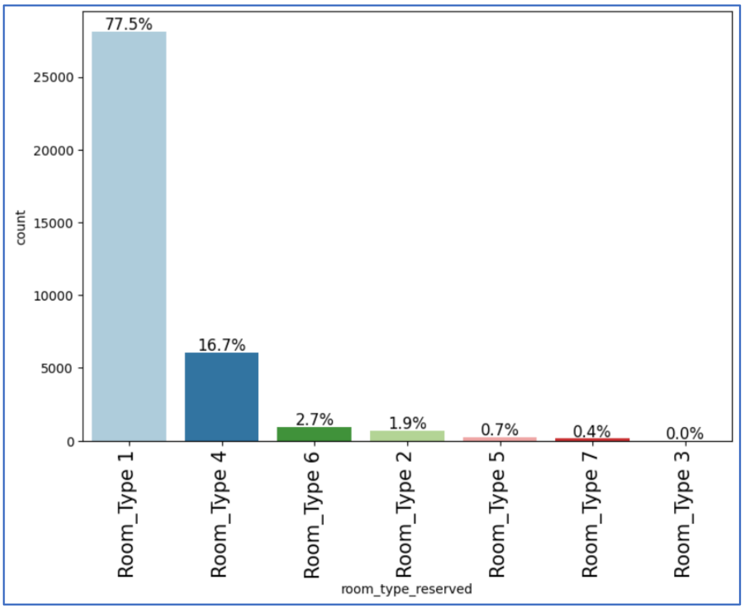


Fig 12. Room type reserved

OBSERVATIONS:

- Approximately 77.7% of customers have selected Room Type 1, while Room Type 4 is the second most popular choice.
- No customers have selected Room Type 3.

4.7.13 Arrival year

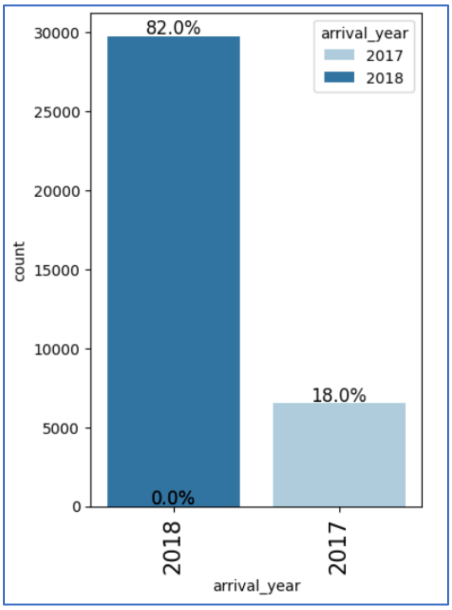


Fig 13. Arrival year

OBSERVATIONS:

- The year 2018 leads in arrival rates with 82%, surpassing 2017.

4.7.14 Arrival month

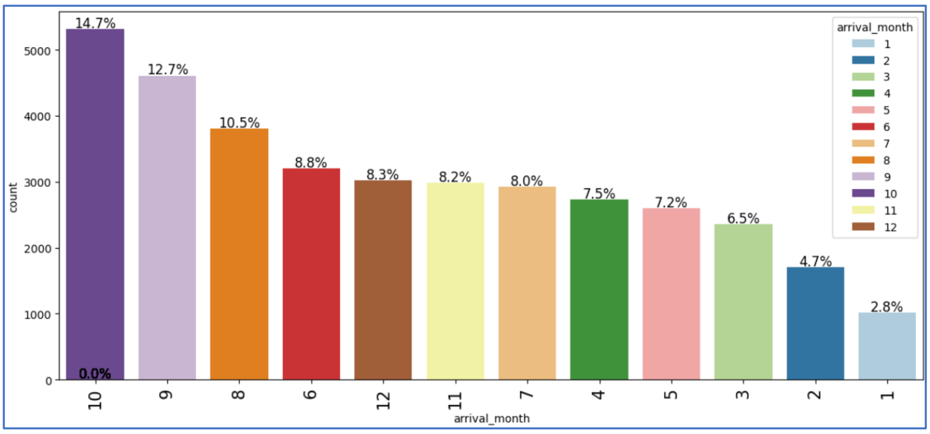


Fig 14. Arrival month

OBSERVATIONS:

- October has the highest customer arrivals at 14.7%, followed by September and august with 12.7% and 10.5% respectively.
- January has the lowest arrival rate, at 2.8%.

greatlearning
Learning for Life

4.7.15 Arrival date

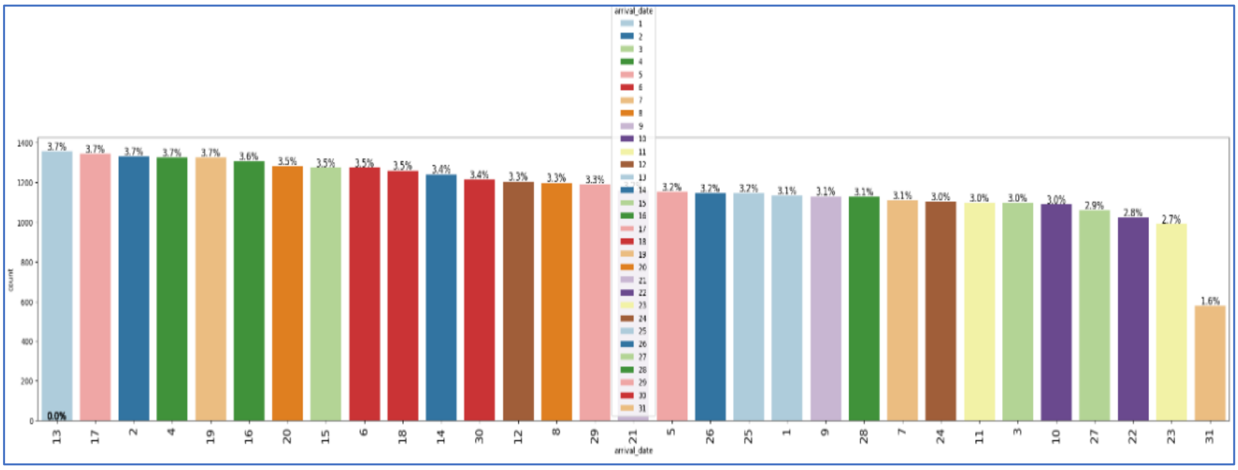


Fig 15. Arrival date

OBSERVATIONS:

- The dates are nearly uniformly distributed, with the 2nd, 4th, 13th, and 17th of the month being the most common, each at 3.7%.
- The 31st is the least common arrival date, at 1.6%, possibly because it does not occur in every month.

4.7.16 Market segment type

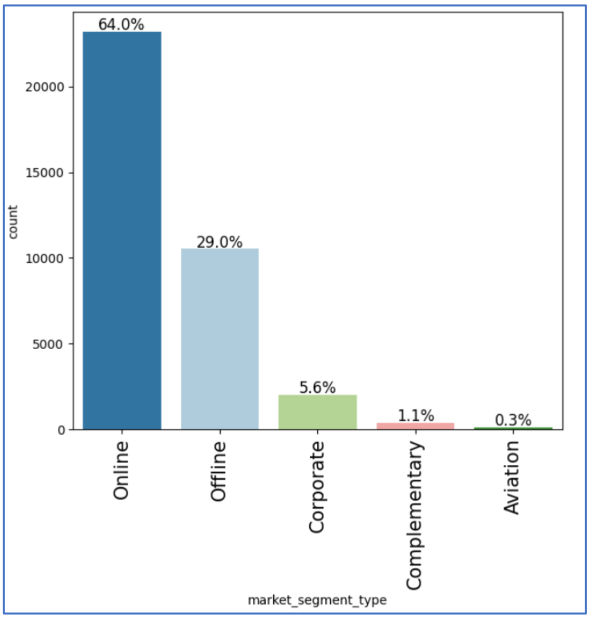


Fig 16. Market segment type

OBSERVATIONS:

- Approximately 64% of the market segment falls under online, while 29% is attributed to offline.
- Corporate, complementary, and aviation segments have the lowest percentages.

4.7.17 Repeated guest

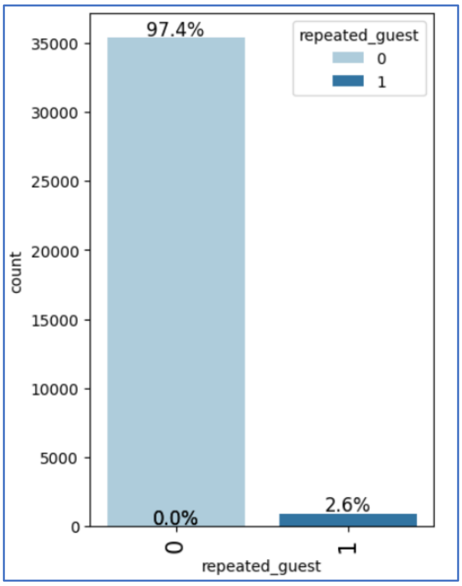


Fig 17. Repeated guest

OBSERVATIONS:

- Nearly 97.4% are first-time guests at the hotel.

4.7.18 Booking status

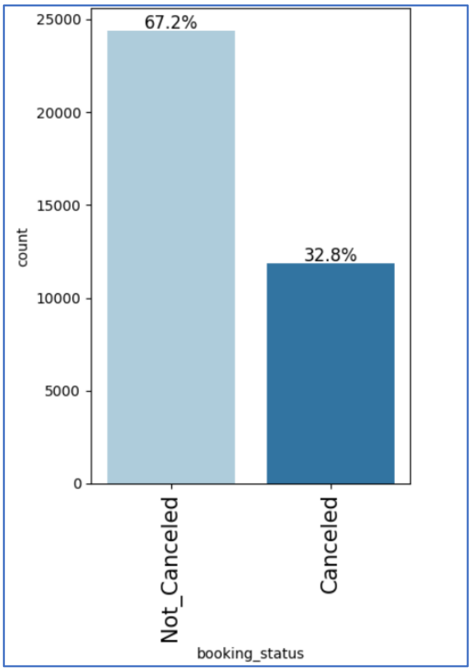


Fig 18. Booking status

OBSERVATIONS:

- 67.2% of the bookings are not cancelled, which is a higher proportion.

4.8 Bivariate analysis

4.8.1 Heat Map

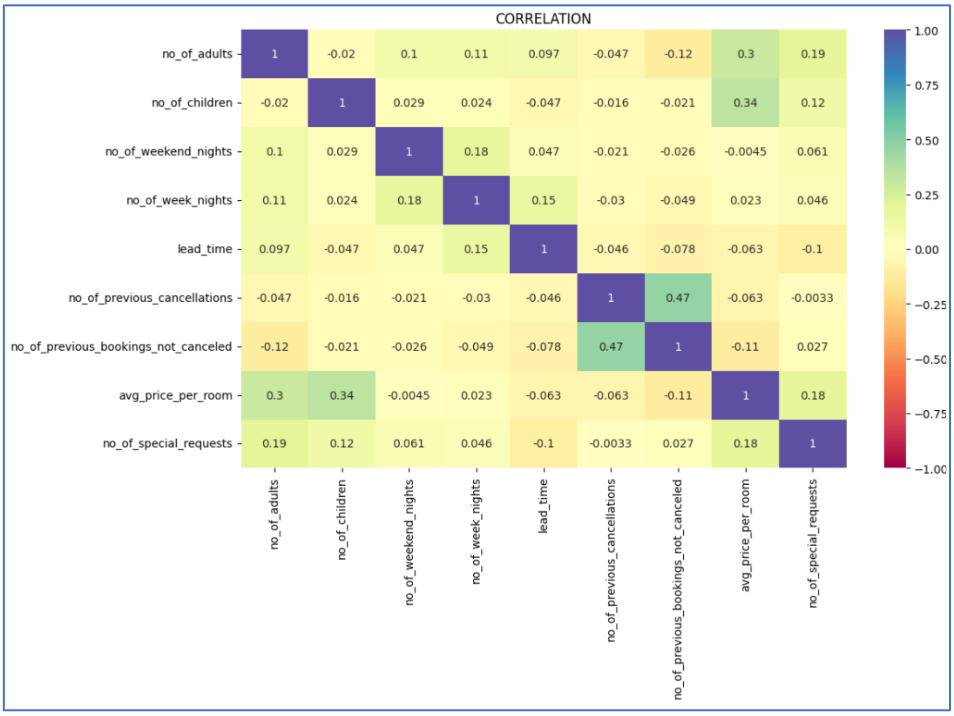


Fig 19. Heat map

OBSERVATIONS:

- A slightly positive correlation, about 0.34 is observed between the number of children and the average price per room.
- A positive correlation of 0.3 is seen between the number of adults and the average price per room.
- About 0.47 positive correlation is observed between number of bookings cancelled and number of booking not cancelled.
- The number of previous cancellations have a strongly negative correlation with the number of children.

4.8.2 Pair Plot for understanding the relationship between numerical variables

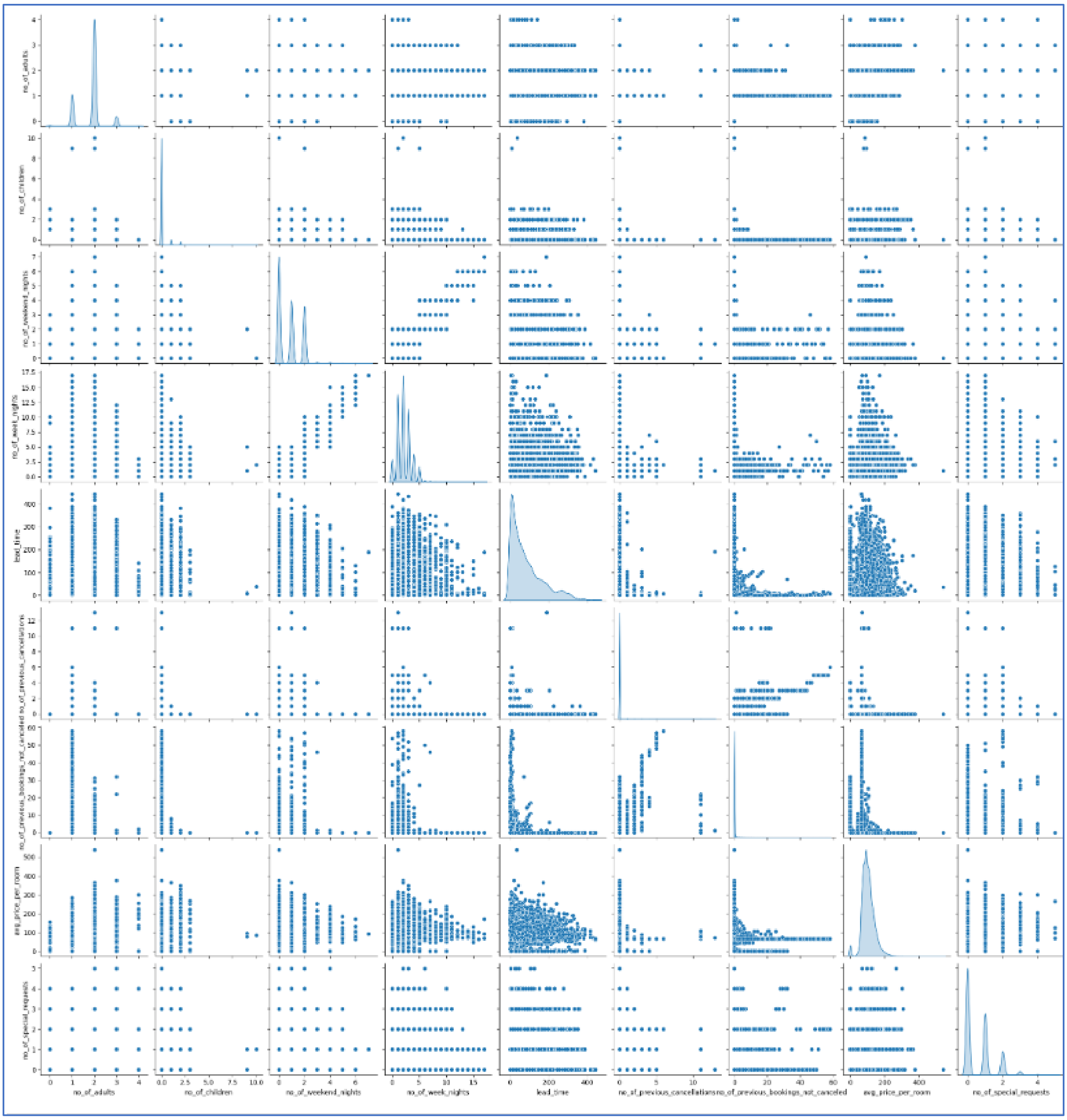


Fig 20. Pair plot

4.8.3 Booking status VS No. of adults

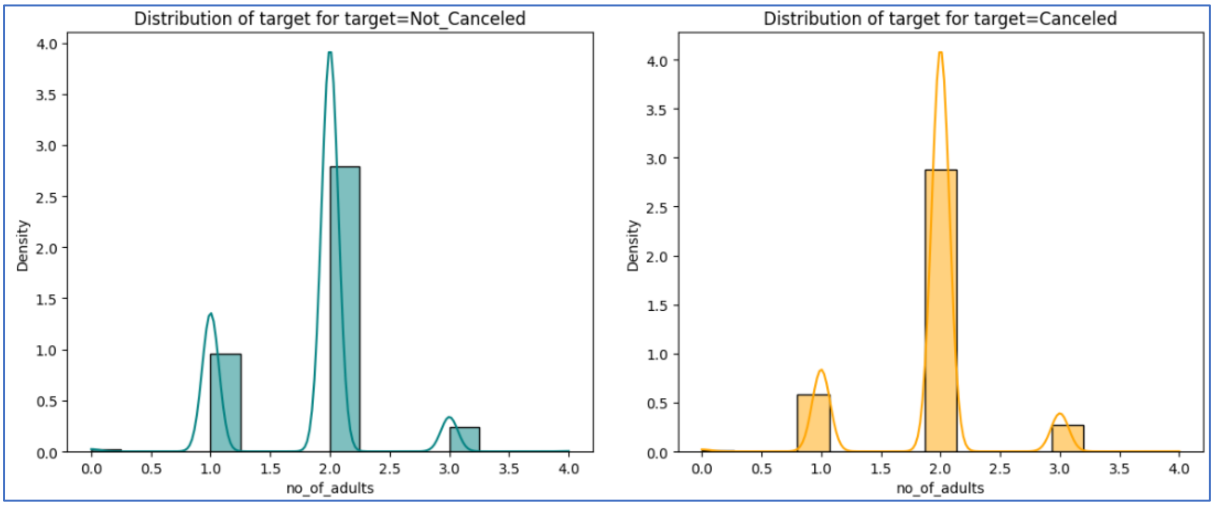


Fig 21. Booking status VS No. of adults

OBSERVATIONS:

- We observe that cancellations decrease when the booking is made for only one adult.

4.8.4 Booking status VS No. of children

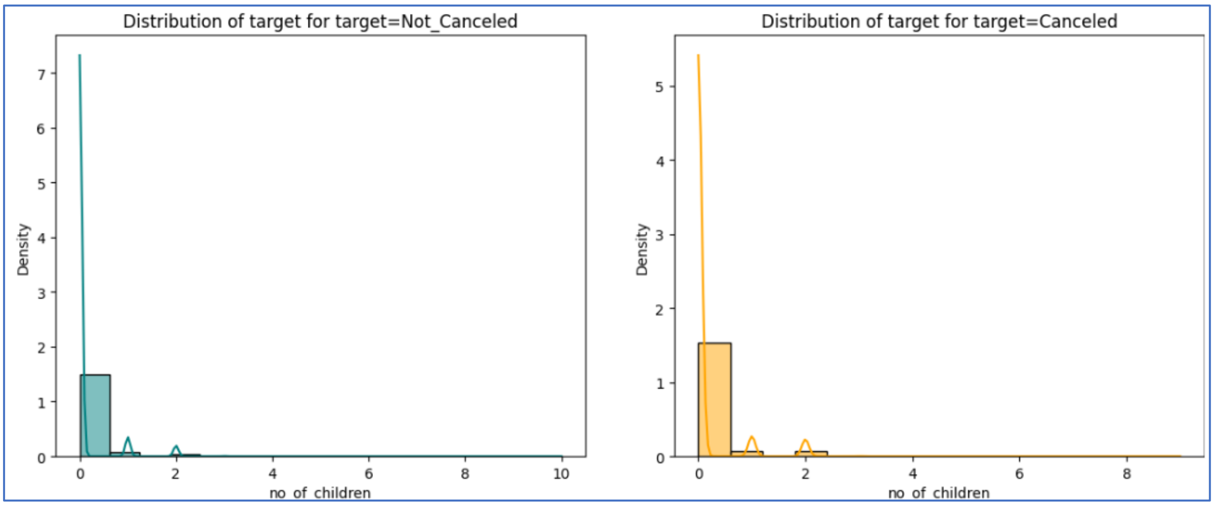


Fig 22. Booking status VS No. of children

OBSERVATIONS:

- We observe that cancellations decrease when there are no children included.

4.8.5 Booking status VS No. of weekend nights

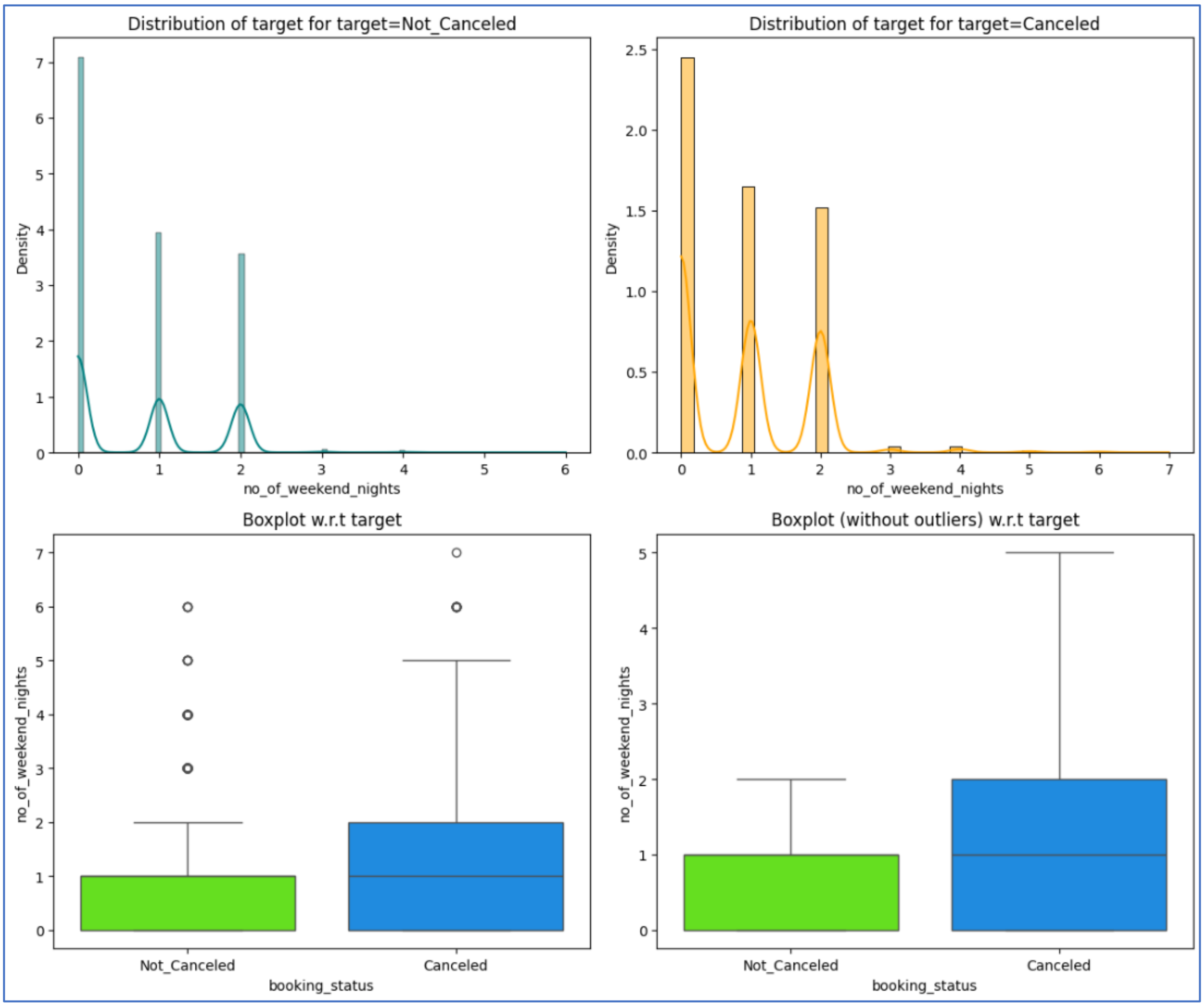


Fig 23. Booking status VS No of weekend nights

OBSERVATIONS:

- Cancellations are more frequent on weekend nights.
- As the number of weekend night bookings rises, so does the cancellation rate.

4.8.6 Booking status VS No. of week nights

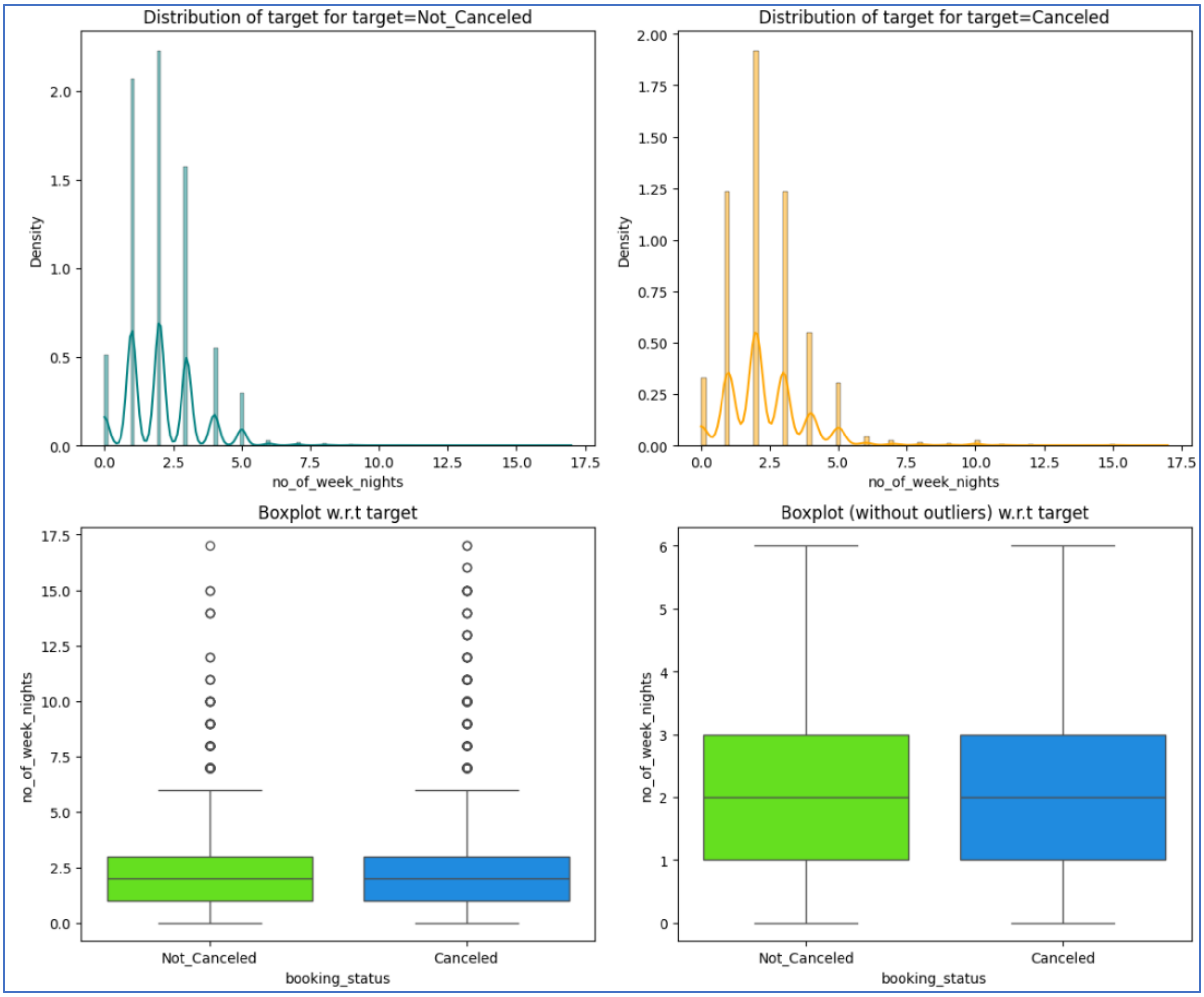


Fig 24. Booking status VS No. of week nights

OBSERVATIONS:

- Cancellations have little impact on weekday bookings.

4.8.7 Booking status VS Lead time

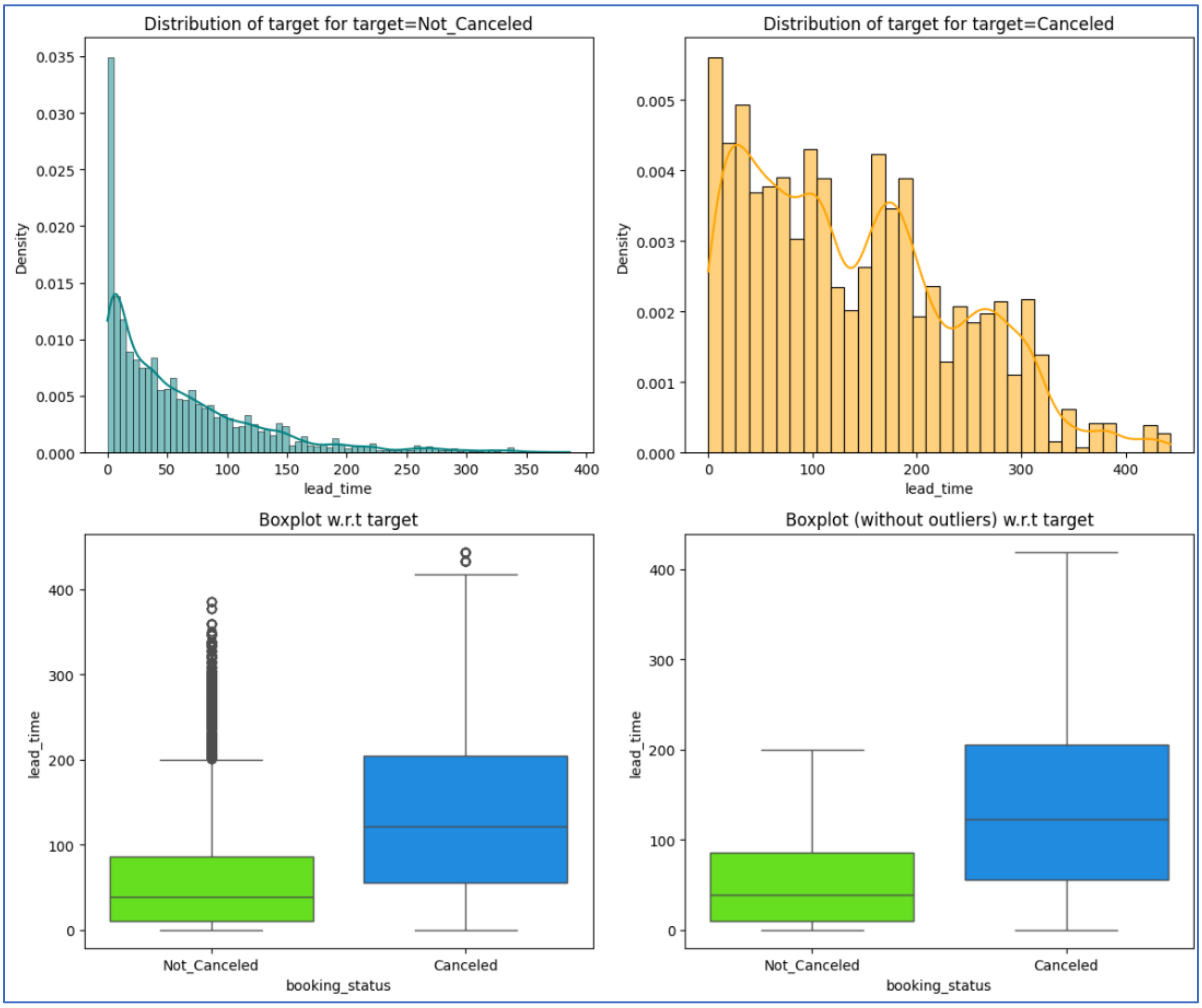


Fig 25. Booking status VS Lead time

OBSERVATIONS:

- Lead time and cancellations are directly proportional; as lead time increases, so does the cancellation rate.

4.8.8 Booking status VS No. of previous cancellations

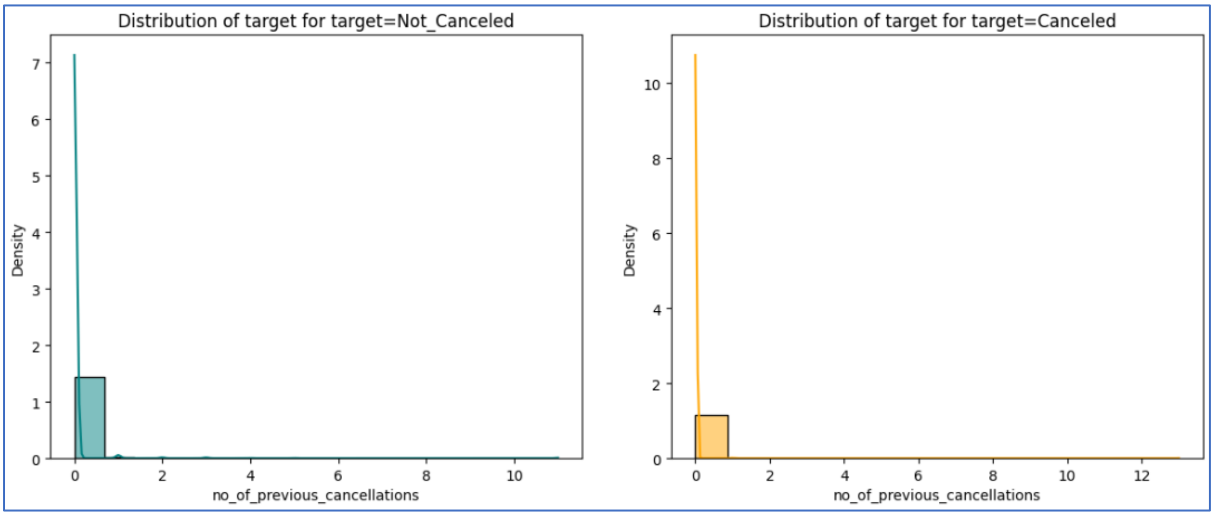


Fig 26. Booking status VS No. of previous cancellations

OBSERVATIONS:

- The number of cancellations is lower among customers who have not previously cancelled bookings.
- The number of previous bookings that were not cancelled is higher than the number of those that were cancelled.

4.8.9 Booking status VS No. of previous booking not cancelled

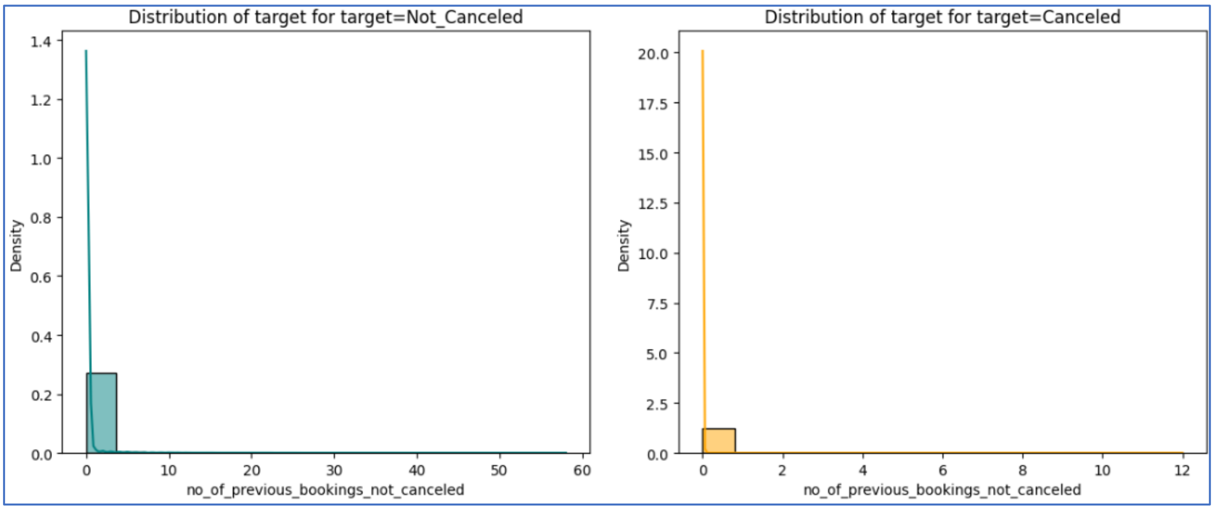


Fig 27. Booking status VS No of previous bookings not cancelled

OBSERVATIONS:

- The number of cancellations is lower among customers who have not previously cancelled bookings.
- The number of previous bookings that were not cancelled is higher than the number of those that were cancelled.

4.8.10 Booking status VS Average price per room

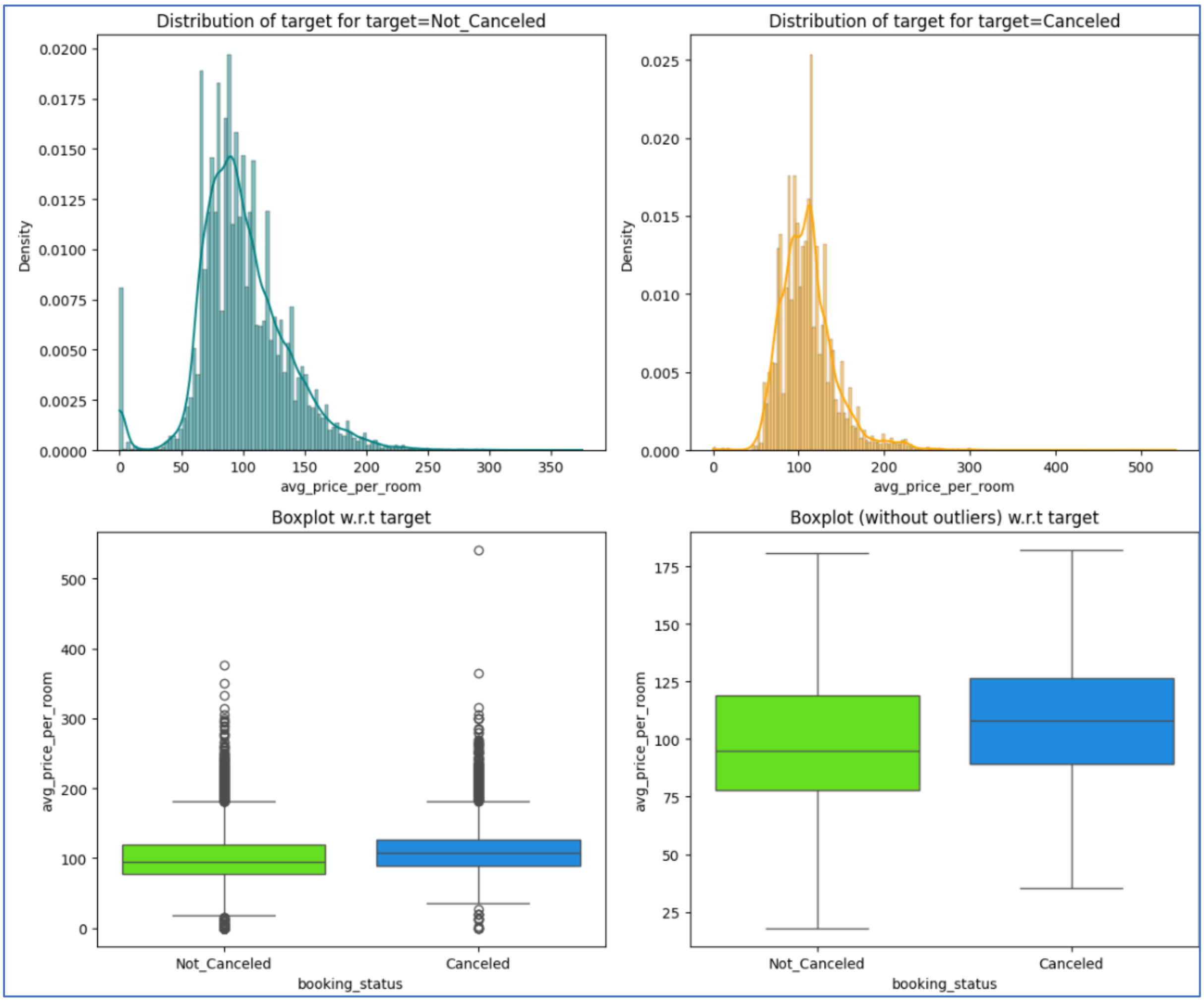


Fig 28. Booking status VS Average price per room

OBSERVATIONS:

- As the average room price rises, the cancellation rate also increases.
- Cancellations are lower for rooms priced at less than 40 euros.

4.8.11 Booking status VS No. of special requests

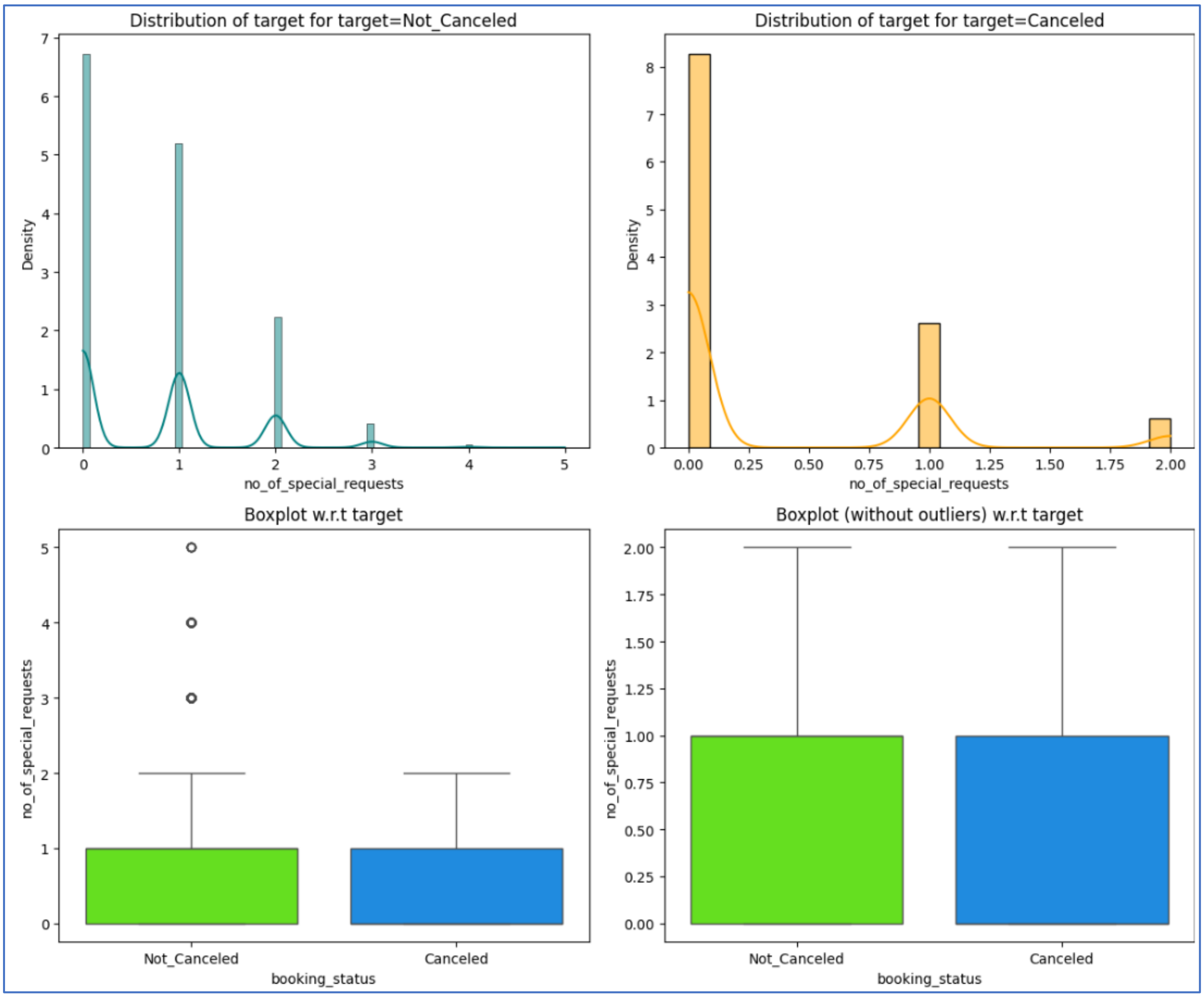


Fig 29. Booking status VS No. of special requests

OBSERVATIONS:

- As the number of requests increases, the cancellation rate decreases.
- However, the number of non-cancelled bookings are higher than those that have been cancelled.

4.8.12 Booking status VS Type of meal plan

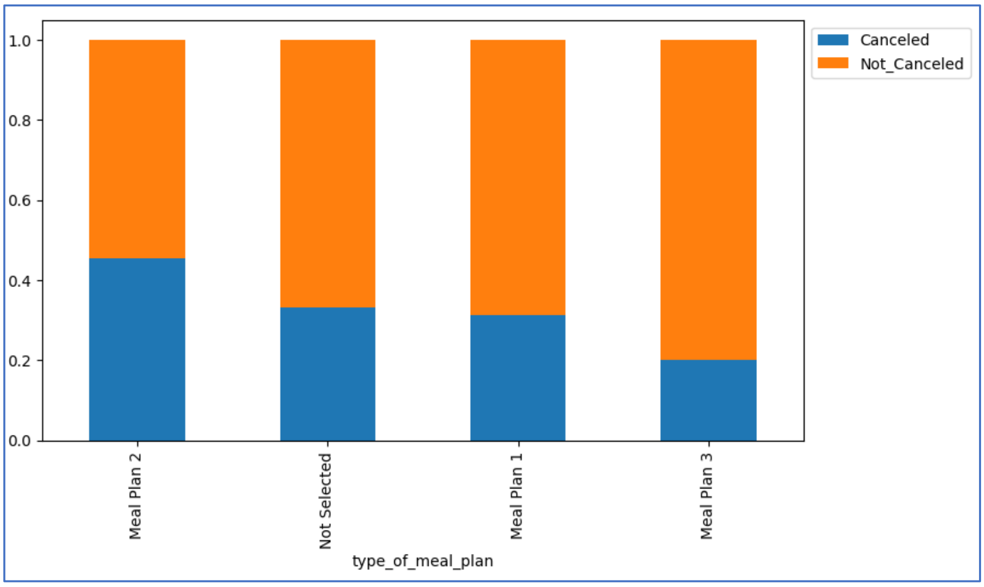


Fig 30. Booking status VS Type of meal plan

OBSERVATIONS:

- Almost 73% of the cancellations occur with customers who chose meal plan 1.
- Cancellations are lower for customers who chose meal plan 3.

4.8.13 Booking status VS Required car parking space

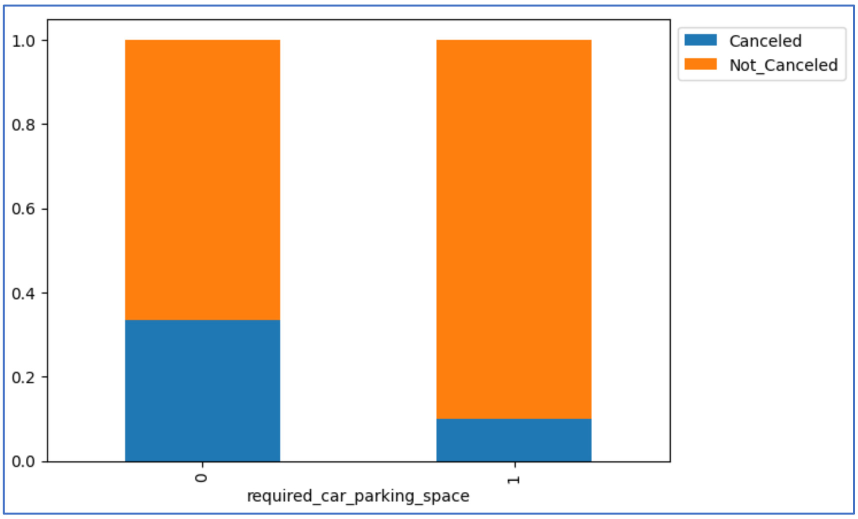


Fig 31. Booking status VS Required car parking space

OBSERVATIONS:

- 99% of customers who do not choose a car parking space end up cancelling their bookings.

4.8.14 Booking status VS Room type reserve

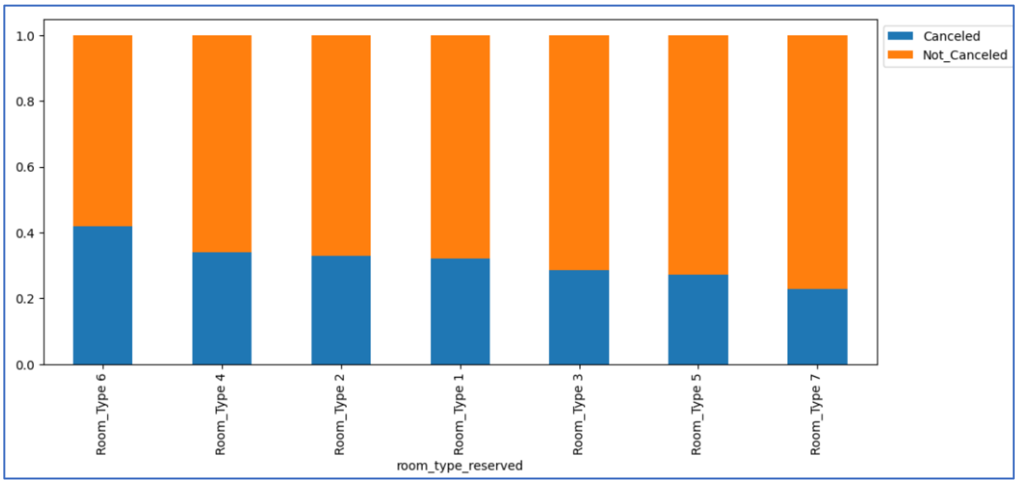


Fig 32. Booking status VS Room type reserved

OBSERVATIONS:

- Customers who selected room type 1 account for 76% of cancellations, followed by room type 4 with 17%, and the fewest cancellations are from room type 3.

4.8.15 Booking status VS Arrival year

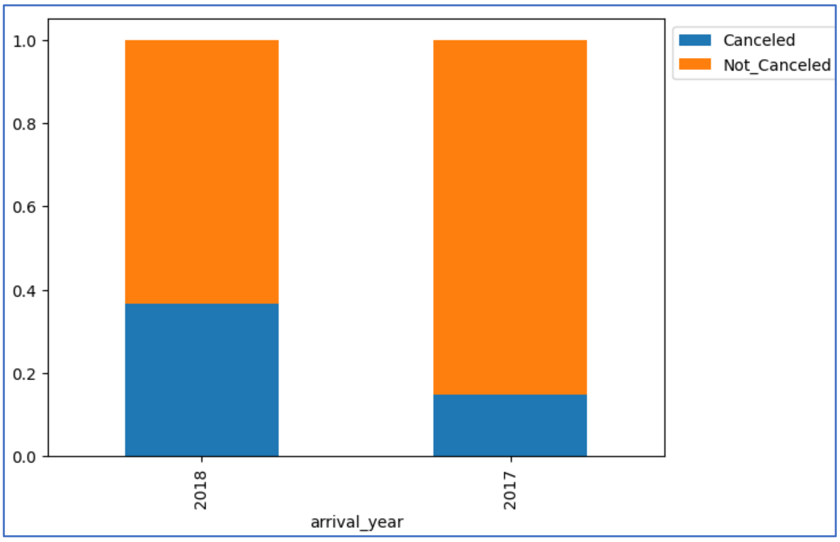


Fig 33. Booking status VS Arrival year

OBSERVATIONS:

- 91% of cancellations occurred in the year 2018.

4.8.16 Booking status VS Arrival month

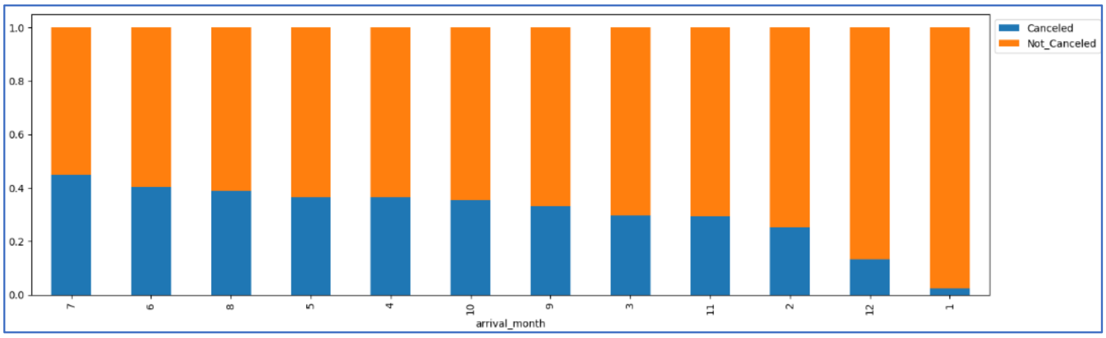


Fig 34. Booking status VS Arrival month

OBSERVATIONS:

- Nearly 15% of cancellations occur in October, followed by 13% in September, with the fewest cancellations occurring in January and December.

4.8.17 Booking status VS Arrival date

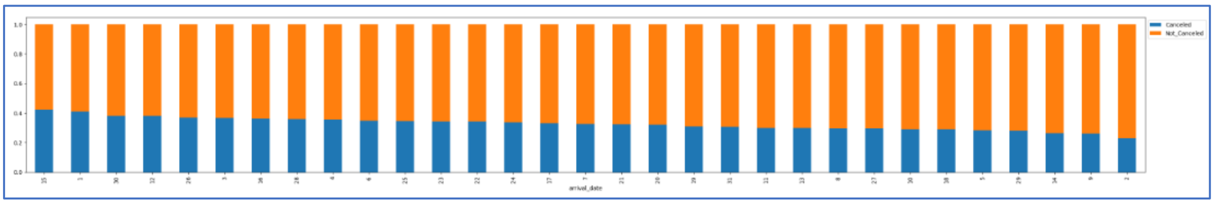


Fig 35. Booking status VS Arrival date

OBSERVATIONS:

- Cancellations are not significantly influenced by the booking date, but the highest number of cancellations occur on the 15th and 4th of the month, while the lowest are seen on the 9th and 31st.

4.8.18 Booking status VS Market segment type

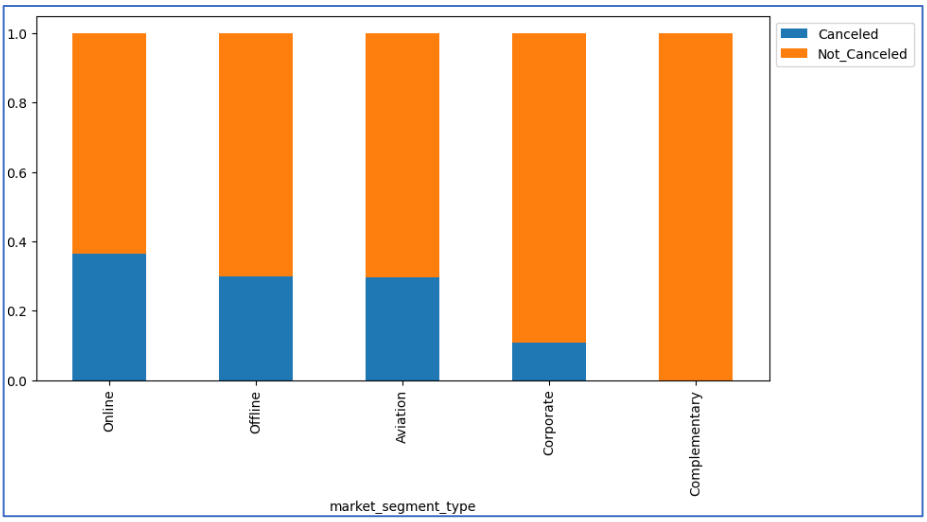


Fig 36. Booking status VS Market segment type

OBSERVATIONS:

- 71% of cancellations occur by customers who have made online bookings.
- There are no cancellations from complementary bookings and least cancellation under aviation.

4.8.19 Booking status VS Repeated guest

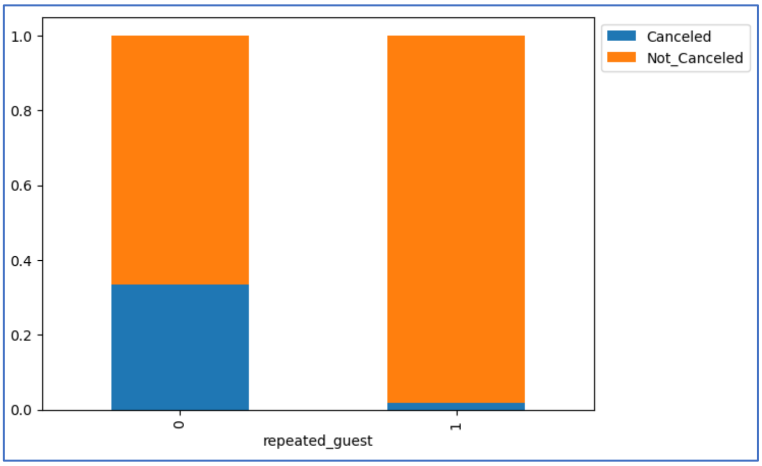


Fig 37. Booking status VS Repeated guest

OBSERVATIONS:

- The number of repeat guests is low, with 99.8% of cancellations occurring among first-time customers.

4.9 Insights based on Exploratory Data Analysis

4.9.1 Based on Univariate analysis:

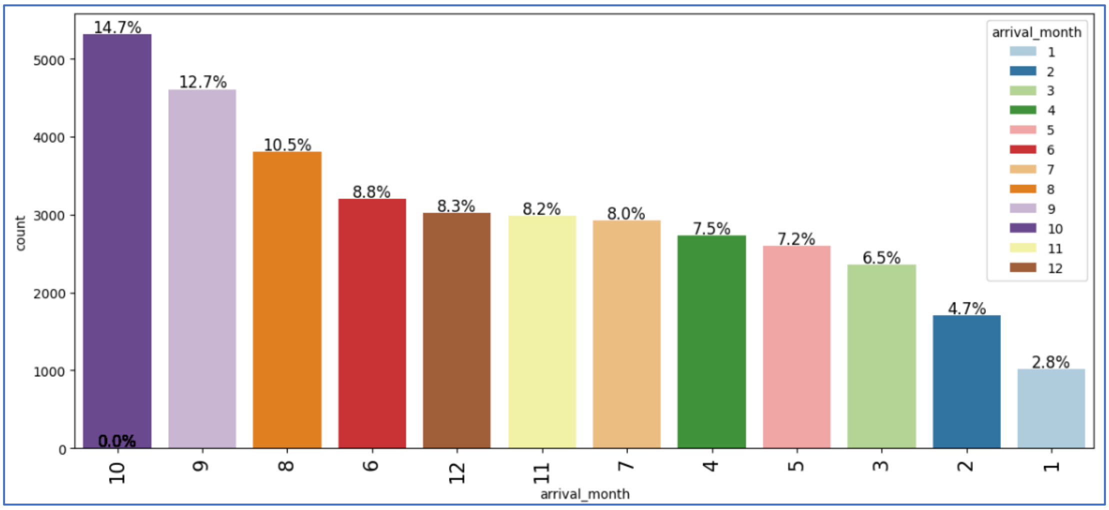
- Most bookings are for 2 adults and typically do not include children.
- Bookings commonly includes 1-2 weekend nights and 2-week nights.
- The average lead time is 90 days, with some bookings extending from 300 to 430 days.
- The previous cancellations range from 0 to 13, while non-cancelled bookings range from 0 to 58.
- The room prices average around 100 euros and are frequently occupied.
- Most customers make only one special request.
- Meal Plan 1 is the most popular (76.7%), followed by Meal Plan 2. Meal Plan 3 is the least chosen.
- Car parking is required by 96.9% of customers.
- Room Type 1 is the most popular (77.7%), with Room Type 3 being least chosen.
- The year 2018 saw the highest arrival rate (82%).
- October had the highest arrivals (14.7%), followed by September (12.7%) and August (10.5%). January had the lowest (2.8%).
- Dates are nearly uniformly distributed, with the 31st being the least common (1.6%).
- 64% of bookings are from the online segment, with offline following.
- 97.4% of guests are first-time visitors.
- 67.2% of bookings are not cancelled, indicating a high proportion of completed bookings.

4.9.2 Based on Bivariate analysis:

- Cancellation rated decrease with bookings for only one adult.
- Fewer cancellations occur when no children are included.
- Cancellations are more frequent with weekend night bookings, where the cancellation rate rises with the number of weekend nights.
- Weekday bookings have very little impact on cancellations.
- The lead time is directly proportional to cancellations. The longer lead times are associated with higher cancellation rates.
- Higher average room prices are associated with increased cancellations. Cancellations are lower for rooms priced below 40 euros.
- Increased requests are associated with decreased cancellations.
- 73% of cancellations are among customers who chose Meal Plan 1. Meal Plan 3 has the lowest cancellation rate.
- 99% of customers who do not choose a car parking space cancel their bookings.
- Room Type 1 has the highest cancellation rate (76%), followed by Room Type 4 (17%). Room Type 3 has the fewest cancellations.
- Highest cancellations are seen in October (15%), followed by September (13%). Fewest in January and December.
- 71% of cancellations are from online bookings. No cancellations from complementary bookings and least cancellations in the aviation segment.
- 99.8% of cancellations are among first-time customers.

5. Questions to be answered as a part of the EDA section

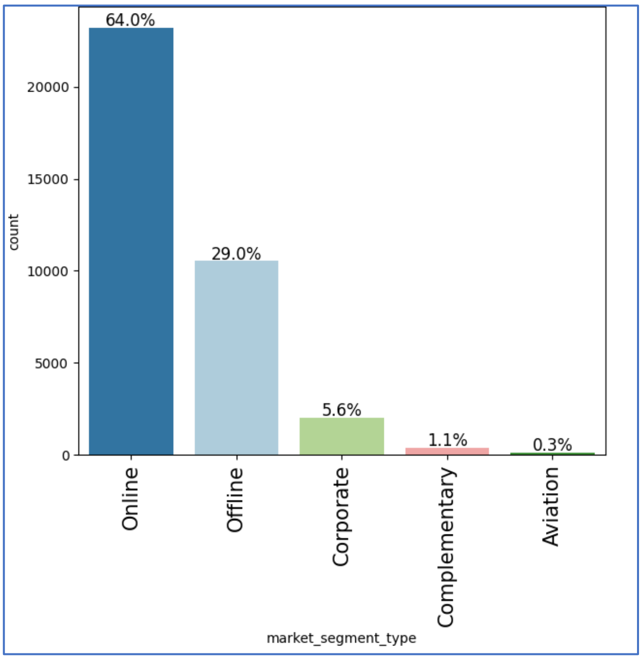
5.1 What are the busiest months in the hotel?



ANSWER:

- October has the highest customer arrivals at 14.7%, followed by September and August with 12.7% and 10.5% respectively.

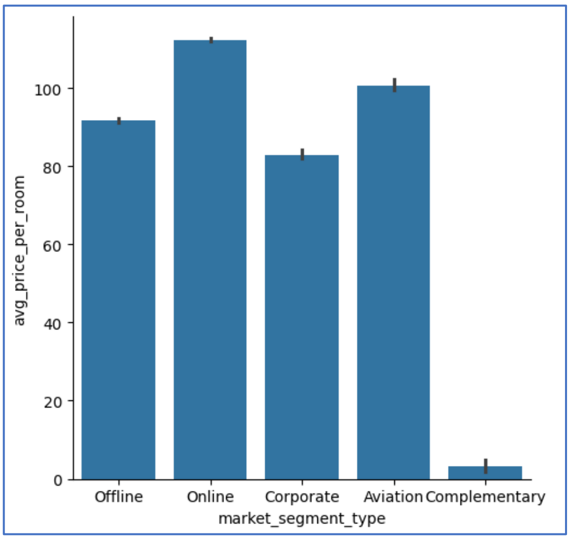
5.2 Which market segment do most of the guests come from?



ANSWER:

- Approximately 64% of the market segment come from online which is the highest, while 29% is attributed to offline.

5.3 Hotel rates are dynamic and change according to demand and customer demographics. What are the differences in room prices in different market segments?

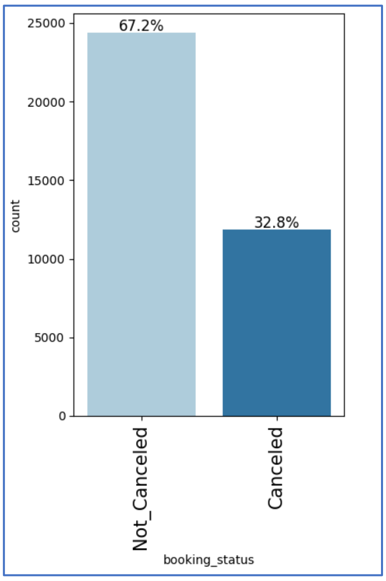


market_segment_type	avg_price_per_room
Aviation	100.704000
Complementary	3.141765
Corporate	82.911740
Offline	91.632679
Online	112.256855

ANSWER:

- The average of average price per room booked online is the highest (112.25 euros), followed by aviation (100.7 euros), offline (91.63 euros), corporate (82.911 euros) and complementary (3.14euros).

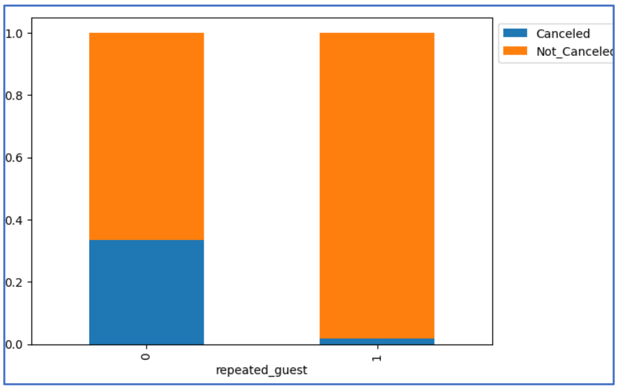
5.4 What percentage of bookings are cancelled?



ANSWER:

- 32.8% of the bookings are cancelled.

5.5 Repeating guests are the guests who stay in the hotel often and are important to brand equity. What percentage of repeating guests cancel?

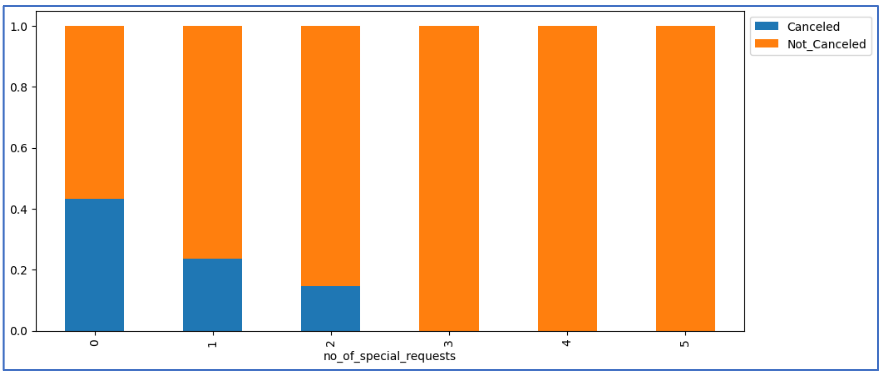



```
repeated_guest  booking_status
0               Not_Canceled      0.664196
                Canceled          0.335804
1               Not_Canceled      0.982796
                Canceled          0.017204
Name: proportion, dtype: float64
```

ANSWER:

- 1.720% of the repeated guests cancel the booking.

5.6 Many guests have special requirements when booking a hotel room. Do these requirements affect booking cancellation?




```
no_of_special_requests  booking_status
0                       Not_Canceled      0.567932
                        Canceled          0.432068
1                       Not_Canceled      0.762332
                        Canceled          0.237668
2                       Not_Canceled      0.854033
                        Canceled          0.145967
3                       Not_Canceled      1.000000
4                       Not_Canceled      1.000000
5                       Not_Canceled      1.000000
Name: proportion, dtype: float64
```

ANSWER:

- An increase in customer special requests is associated with a decrease in booking cancellations.
- Customers with no special requests had a cancellation rate of 43%, while those with one special request had a 23% cancellation rate, and those with two special requests had a 14.5% cancellation rate.
- Customers who make between 3 and 5 requests have no cancellations.

6. Data Preprocessing

6.1 Duplicate value check

- No duplicate values are found in the data set.

6.2 Missing value treatment

- No missing values or null values are found in the data set.

6.3 Outlier detection

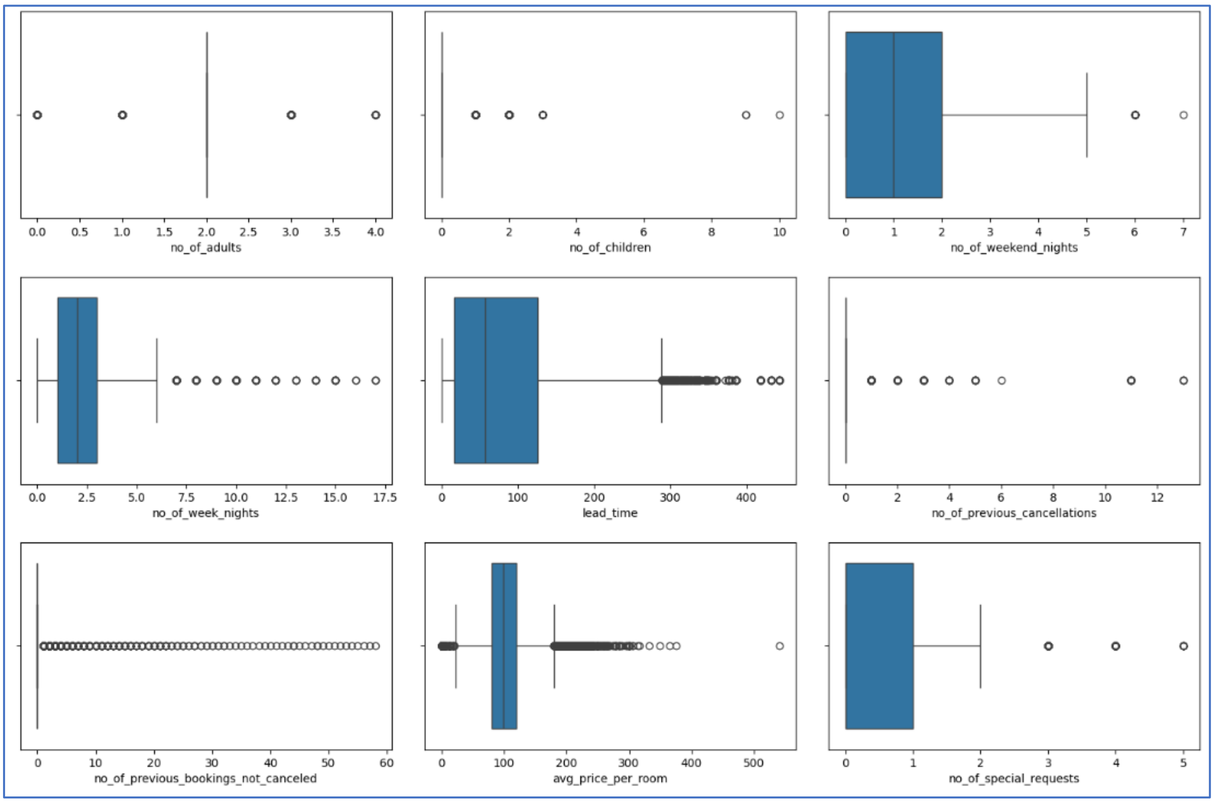


Fig 38. Outlier detection

OBSERVATIONS:

- There are a few outliers in the data.
- The outliers in number of adults, number of children, number of weekend nights, number of week nights, lead time, number of previous cancellations, number of previous bookings not cancelled and number of special requests, add value to the data.
- The average price per room costs is 0 for some entries, however they are found to be from complementary and online bookings, which could be due to rewards and offers provided by the hotel.
- Hence, we are not treating any outliers, as they all are data that add value.

6.4 Feature Engineering

- The Booking_ID column has unique values, which are not important to data set. Hence the column is dropped from the data set.
- The booking_status column has Canceled and Non_Canceled entries, which are converted into binary numbers, 1 and 0 respectively.
- The categorical columns are encoded to dummy variables using one-hot encoding.

6.5 Data preparation

- The data set has been split into training and test set with a split of 70% and 30% respectively.

```
Shape of Training set :  (25392, 27)
Shape of test set :  (10883, 27)
Percentage of classes in training set:
booking_status
0    0.670644
1    0.329356
Name: proportion, dtype: float64
Percentage of classes in test set:
booking_status
0    0.676376
1    0.323624
Name: proportion, dtype: float64
```

- Around 967% of observations belongs to class 0 (Not_Canceled) and 32% observations belongs to class 1 (Canceled), and this is preserved in the train and test sets.

6.6 Model Building

The model can make wrong predictions as:

- Predicting cancellation will not happen but in reality, the cancellation will happen (False negative, FN)

- Predicting cancellation will happen but in reality, the cancellation will not happen (False positive, FP)

The INN Hotel group would want to reduce false negatives, which can be done by maximizing the Recall. Greater the recall lesser the chances of false negatives.

7. Logistic Regression (with statsmodels library)

7.1 Logistic regression result

The logistic regression model obtained using the train data set is as follows:

Logit Regression Results

Dep. Variable:	booking_status	No. Observations:	25392
Model:	Logit	Df Residuals:	25364
Method:	MLE	Df Model:	27
Date:	Sat, 10 Aug 2024	Pseudo R-squ.:	0.3293
Time:	10:29:44	Log-Likelihood:	-10793.
converged:	False	LL-Null:	-16091.
Covariance Type:	nonrobust	LLR p-value:	0.000

	coef	std err	z	P>|z|	[0.025	0.975]
const	-924.5923	120.817	-7.653	0.000	-1161.390	-687.795
no_of_adults	0.1135	0.038	3.017	0.003	0.040	0.187
no_of_children	0.1563	0.057	2.732	0.006	0.044	0.268
no_of_weekend_nights	0.1068	0.020	5.398	0.000	0.068	0.146
no_of_week_nights	0.0398	0.012	3.239	0.001	0.016	0.064
required_car_parking_space	-1.5939	0.138	-11.561	0.000	-1.864	-1.324
lead_time	0.0157	0.000	58.868	0.000	0.015	0.016
arrival_year	0.4570	0.060	7.633	0.000	0.340	0.574
arrival_month	-0.0415	0.006	-6.418	0.000	-0.054	-0.029
arrival_date	0.0005	0.002	0.252	0.801	-0.003	0.004
repeated_guest	-2.3469	0.617	-3.805	0.000	-3.556	-1.138
no_of_previous_cancellations	0.2664	0.086	3.108	0.002	0.098	0.434
no_of_previous_bookings_not_canceled	-0.1727	0.153	-1.131	0.258	-0.472	0.127
avg_price_per_room	0.0188	0.001	25.404	0.000	0.017	0.020
no_of_special_requests	-1.4690	0.030	-48.790	0.000	-1.528	-1.410
type_of_meal_plan_Meal Plan 2	0.1768	0.067	2.654	0.008	0.046	0.307
type_of_meal_plan_Meal Plan 3	17.8379	5057.771	0.004	0.997	-9895.212	9930.887
type_of_meal_plan_Not Selected	0.2782	0.053	5.245	0.000	0.174	0.382
room_type_reserved_Room_Type 2	-0.3610	0.131	-2.761	0.006	-0.617	-0.105
room_type_reserved_Room_Type 3	-0.0009	1.310	-0.001	0.999	-2.569	2.567
room_type_reserved_Room_Type 4	-0.2821	0.053	-5.305	0.000	-0.386	-0.178
room_type_reserved_Room_Type 5	-0.7176	0.209	-3.432	0.001	-1.127	-0.308
room_type_reserved_Room_Type 6	-0.9456	0.147	-6.434	0.000	-1.234	-0.658
room_type_reserved_Room_Type 7	-1.3964	0.293	-4.767	0.000	-1.971	-0.822
market_segment_type_Complementary	-41.8798	8.42e+05	-4.98e-05	1.000	-1.65e+06	1.65e+06
market_segment_type_Corporate	-1.1935	0.266	-4.487	0.000	-1.715	-0.672
market_segment_type_Offline	-2.1955	0.255	-8.625	0.000	-2.694	-1.697
market_segment_type_Online	-0.3990	0.251	-1.588	0.112	-0.891	0.093

Table 3. Logistic regression result

OBSERVATIONS:

- The negative values of the coefficient shows that probability of a customer cancelling the booking decreases with the increase of corresponding attribute value. The attributes here with negative coefficients include required_car_parking_space, arrival_month, repeated_guest, no_of_previous_bookings_not_canceled, no_of_special_requests, room_type_reserved and market_segment_type.
- The positive values of the coefficient shows that probability of a customer cancelling the booking increases with the increase of corresponding attribute value. The attributes here with positive coefficients include no_of_adults, no_of_children, no_of_weekend_night, no_of_week_nigh, lead_time, arrival_year, arrival_date, no_of_previous_cancellations, avg_price_per_room and type_of_meal_plan.
- The p-value of a variable indicates if the variable is significant or not. If we consider the significance level to be 0.05 (5%), then any variable with a p-value less than 0.05 would be considered significant.
- However these variables might contain multicollinearity, which will affect the p-value.
- We will have to remove multicollinearity.

7.1.2 Test multicollinearity by checking variance inflation factor (VIF)

```
Series before feature selection:

const                                  3.946816e+07
no_of_adults                           1.348154e+00
no_of_children                         1.978229e+00
no_of_weekend_nights                   1.069475e+00
no_of_week_nights                      1.095667e+00
required_car_parking_space             1.039928e+00
lead_time                              1.394914e+00
arrival_year                           1.430830e+00
arrival_month                          1.275673e+00
arrival_date                           1.006738e+00
repeated_guest                         1.783516e+00
no_of_previous_cancellations           1.395689e+00
no_of_previous_bookings_not_canceled   1.651986e+00
avg_price_per_room                     2.050421e+00
no_of_special_requests                 1.247278e+00
type_of_meal_plan_Meal Plan 2          1.271851e+00
type_of_meal_plan_Meal Plan 3          1.025216e+00
type_of_meal_plan_Not Selected         1.272183e+00
room_type_reserved_Room_Type 2         1.101438e+00
room_type_reserved_Room_Type 3         1.003302e+00
room_type_reserved_Room_Type 4         1.361515e+00
room_type_reserved_Room_Type 5         1.027810e+00
room_type_reserved_Room_Type 6         1.973072e+00
room_type_reserved_Room_Type 7         1.115123e+00
market_segment_type_Complementary      4.500109e+00
market_segment_type_Corporate          1.692844e+01
market_segment_type_Offline            6.411392e+01
market_segment_type_Online             7.117643e+01
dtype: float64
```

Table 4. VIF result 1

OBSERVATIONS:

- Three columns, namely market_segment_type_Offline, market_segment_type_Corporate and market_segment_type_Online have VIF greater than 5.
- Hence, we are dropping market_segment_type_Offline which has a high VIF first, and checking the VIF values again.

greatlearning
Learning for Life

7.1.2.1 Dropping market_segment_type_Offline

```
Series before feature selection:

const                                    3.935697e+07
no_of_adults                             1.332814e+00
no_of_children                           1.977870e+00
no_of_weekend_nights                     1.068494e+00
no_of_week_nights                        1.094730e+00
required_car_parking_space               1.039643e+00
lead_time                                1.387302e+00
arrival_year                             1.427201e+00
arrival_month                            1.274917e+00
arrival_date                             1.006708e+00
repeated_guest                           1.780535e+00
no_of_previous_cancellations             1.395535e+00
no_of_previous_bookings_not_canceled     1.651657e+00
avg_price_per_room                       2.050071e+00
no_of_special_requests                   1.246880e+00
type_of_meal_plan_Meal Plan 2            1.271450e+00
type_of_meal_plan_Meal Plan 3            1.025216e+00
type_of_meal_plan_Not Selected           1.272123e+00
room_type_reserved_Room_Type 2           1.101429e+00
room_type_reserved_Room_Type 3           1.003301e+00
room_type_reserved_Room_Type 4           1.352330e+00
room_type_reserved_Room_Type 5           1.027776e+00
room_type_reserved_Room_Type 6           1.972844e+00
room_type_reserved_Room_Type 7           1.115087e+00
market_segment_type_Complementary        1.314313e+00
market_segment_type_Corporate            1.528363e+00
market_segment_type_Online               1.773383e+00
dtype: float64
```

Table 5. VIF result 2

```
                           Logit Regression Results
==============================================================================
Dep. Variable:         booking_status   No. Observations:                25392
Model:                          Logit   Df Residuals:                    25365
Method:                           MLE   Df Model:                           26
Date:                Sat, 10 Aug 2024   Pseudo R-squ.:                  0.3275
Time:                        15:59:41   Log-Likelihood:                -10821.
converged:                      False   LL-Null:                       -16091.
Covariance Type:            nonrobust   LLR p-value:                     0.000
==========================================================================================================
                                           coef    std err          z      P>|z|      [0.025      0.975]
----------------------------------------------------------------------------------------------------------
const                                 -965.8027    120.408     -8.021      0.000   -1201.798    -729.808
no_of_adults                             0.0886      0.037      2.368      0.018       0.015       0.162
no_of_children                           0.1525      0.057      2.668      0.008       0.040       0.264
no_of_weekend_nights                     0.1120      0.020      5.665      0.000       0.073       0.151
no_of_week_nights                        0.0438      0.012      3.564      0.000       0.020       0.068
required_car_parking_space              -1.5755      0.138    -11.436      0.000      -1.846      -1.306
lead_time                                0.0155      0.000     58.696      0.000       0.015       0.016
arrival_year                             0.4763      0.060      7.982      0.000       0.359       0.593
arrival_month                           -0.0401      0.006     -6.210      0.000      -0.053      -0.027
arrival_date                             0.0006      0.002      0.284      0.776      -0.003       0.004
repeated_guest                          -2.2463      0.620     -3.626      0.000      -3.461      -1.032
no_of_previous_cancellations             0.2568      0.086      2.997      0.003       0.089       0.425
no_of_previous_bookings_not_canceled    -0.1730      0.151     -1.146      0.252      -0.469       0.123
avg_price_per_room                       0.0187      0.001     25.363      0.000       0.017       0.020
no_of_special_requests                  -1.4649      0.030    -48.791      0.000      -1.524      -1.406
type_of_meal_plan_Meal Plan 2            0.1687      0.066      2.542      0.011       0.039       0.299
type_of_meal_plan_Meal Plan 3           26.1700   3.28e+05   7.98e-05      1.000   -6.43e+05    6.43e+05
type_of_meal_plan_Not Selected           0.2800      0.053      5.289      0.000       0.176       0.384
room_type_reserved_Room_Type 2          -0.3558      0.130     -2.728      0.006      -0.612      -0.100
room_type_reserved_Room_Type 3          -0.0036      1.302     -0.003      0.998      -2.555       2.548
room_type_reserved_Room_Type 4          -0.2531      0.053     -4.766      0.000      -0.357      -0.149
room_type_reserved_Room_Type 5          -0.7212      0.208     -3.462      0.001      -1.129      -0.313
room_type_reserved_Room_Type 6          -0.9287      0.147     -6.330      0.000      -1.216      -0.641
room_type_reserved_Room_Type 7          -1.3783      0.292     -4.713      0.000      -1.951      -0.805
market_segment_type_Complementary      -67.9385   1.72e+10  -3.96e-09      1.000   -3.37e+10    3.37e+10
market_segment_type_Corporate            0.9450      0.106      8.898      0.000       0.737       1.153
market_segment_type_Online               1.7488      0.051     33.985      0.000       1.648       1.850
==========================================================================================================
```

Table 6. Logistic regression result 1

OBSERVATIONS:

- By dropping one column, market_segment_type_Offline, the VIF of all the variables have been adjusted to lesser than 2.
- Hence, we have removed multicollinearity from the data.
- It is necessary to remove the insignificant features (p-value>0.05).

greatlearning
Learning for Life

7.1.2.2 Dropping market segment type complementary

Dropping the column with p = 1

Logit Regression Results

Dep. Variable:	booking_status	No. Observations:	25392
Model:	Logit	Df Residuals:	25366
Method:	MLE	Df Model:	25
Date:	Sat, 10 Aug 2024	Pseudo R-squ.:	0.3273
Time:	16:03:22	Log-Likelihood:	-10825.
converged:	True	LL-Null:	-16091.
Covariance Type:	nonrobust	LLR p-value:	0.000

	coef	std err	z	P>|z|	[0.025	0.975]
const	-958.6646	120.330	-7.967	0.000	-1194.508	-722.821
no_of_adults	0.0899	0.037	2.404	0.016	0.017	0.163
no_of_children	0.1518	0.057	2.656	0.008	0.040	0.264
no_of_weekend_nights	0.1117	0.020	5.649	0.000	0.073	0.150
no_of_week_nights	0.0444	0.012	3.610	0.000	0.020	0.068
required_car_parking_space	-1.5777	0.138	-11.453	0.000	-1.848	-1.308
lead_time	0.0155	0.000	58.793	0.000	0.015	0.016
arrival_year	0.4728	0.060	7.928	0.000	0.356	0.590
arrival_month	-0.0405	0.006	-6.281	0.000	-0.053	-0.028
arrival_date	0.0005	0.002	0.249	0.804	-0.003	0.004
repeated_guest	-2.2492	0.619	-3.632	0.000	-3.463	-1.036
no_of_previous_cancellations	0.2573	0.086	3.003	0.003	0.089	0.425
no_of_previous_bookings_not_canceled	-0.1730	0.151	-1.146	0.252	-0.469	0.123
avg_price_per_room	0.0188	0.001	25.631	0.000	0.017	0.020
no_of_special_requests	-1.4652	0.030	-48.808	0.000	-1.524	-1.406
type_of_meal_plan_Meal Plan 2	0.1648	0.066	2.484	0.013	0.035	0.295
type_of_meal_plan_Meal Plan 3	3.0535	2.431	1.256	0.209	-1.711	7.818
type_of_meal_plan_Not Selected	0.2816	0.053	5.320	0.000	0.178	0.385
room_type_reserved_Room_Type 2	-0.3544	0.130	-2.717	0.007	-0.610	-0.099
room_type_reserved_Room_Type 3	-0.0231	1.289	-0.018	0.986	-2.549	2.502
room_type_reserved_Room_Type 4	-0.2565	0.053	-4.832	0.000	-0.361	-0.152
room_type_reserved_Room_Type 5	-0.7255	0.208	-3.485	0.000	-1.134	-0.317
room_type_reserved_Room_Type 6	-0.9374	0.147	-6.391	0.000	-1.225	-0.650
room_type_reserved_Room_Type 7	-1.3976	0.292	-4.786	0.000	-1.970	-0.825
market_segment_type_Corporate	0.9484	0.106	8.932	0.000	0.740	1.157
market_segment_type_Online	1.7502	0.051	34.019	0.000	1.649	1.851

Table 7. Logistic regression result 2

7.1.2.3 Dropping room type reserved room type 3

Dropping the column with p = 0.986

Logit Regression Results

Dep. Variable:	booking_status	No. Observations:	25392
Model:	Logit	Df Residuals:	25367
Method:	MLE	Df Model:	24
Date:	Sat, 10 Aug 2024	Pseudo R-squ.:	0.3273
Time:	16:05:13	Log-Likelihood:	-10825.
converged:	True	LL-Null:	-16091.
Covariance Type:	nonrobust	LLR p-value:	0.000

	coef	std err	z	P>|z|	[0.025	0.975]
const	-958.6544	120.329	-7.967	0.000	-1194.495	-722.814
no_of_adults	0.0899	0.037	2.404	0.016	0.017	0.163
no_of_children	0.1518	0.057	2.656	0.008	0.040	0.264
no_of_weekend_nights	0.1117	0.020	5.649	0.000	0.073	0.150
no_of_week_nights	0.0444	0.012	3.610	0.000	0.020	0.068
required_car_parking_space	-1.5777	0.138	-11.453	0.000	-1.848	-1.308
lead_time	0.0155	0.000	58.793	0.000	0.015	0.016
arrival_year	0.4728	0.060	7.928	0.000	0.356	0.590
arrival_month	-0.0405	0.006	-6.281	0.000	-0.053	-0.028
arrival_date	0.0005	0.002	0.249	0.803	-0.003	0.004
repeated_guest	-2.2492	0.619	-3.632	0.000	-3.463	-1.036
no_of_previous_cancellations	0.2573	0.086	3.003	0.003	0.089	0.425
no_of_previous_bookings_not_canceled	-0.1730	0.151	-1.146	0.252	-0.469	0.123
avg_price_per_room	0.0188	0.001	25.631	0.000	0.017	0.020
no_of_special_requests	-1.4652	0.030	-48.810	0.000	-1.524	-1.406
type_of_meal_plan_Meal Plan 2	0.1648	0.066	2.484	0.013	0.035	0.295
type_of_meal_plan_Meal Plan 3	3.0535	2.431	1.256	0.209	-1.711	7.818
type_of_meal_plan_Not Selected	0.2816	0.053	5.320	0.000	0.178	0.385
room_type_reserved_Room_Type 2	-0.3544	0.130	-2.717	0.007	-0.610	-0.099
room_type_reserved_Room_Type 4	-0.2565	0.053	-4.832	0.000	-0.361	-0.152
room_type_reserved_Room_Type 5	-0.7255	0.208	-3.485	0.000	-1.134	-0.317
room_type_reserved_Room_Type 6	-0.9374	0.147	-6.391	0.000	-1.225	-0.650
room_type_reserved_Room_Type 7	-1.3976	0.292	-4.786	0.000	-1.970	-0.825
market_segment_type_Corporate	0.9484	0.106	8.932	0.000	0.740	1.157
market_segment_type_Online	1.7502	0.051	34.019	0.000	1.649	1.851

Table 8. Logistic regression result 3

7.1.2.4 Dropping meal plan 3

Dropping the column with p = 0.209

Logit Regression Results

Dep. Variable:	booking_status	No. Observations:	25392
Model:	Logit	Df Residuals:	25368
Method:	MLE	Df Model:	23
Date:	Sat, 10 Aug 2024	Pseudo R-squ.:	0.3272
Time:	16:05:59	Log-Likelihood:	-10826.
converged:	True	LL-Null:	-16091.
Covariance Type:	nonrobust	LLR p-value:	0.000

	coef	std err	z	P>|z|	[0.025	0.975]
const	-955.0519	120.281	-7.940	0.000	-1190.799	-719.305
no_of_adults	0.0899	0.037	2.404	0.016	0.017	0.163
no_of_children	0.1514	0.057	2.648	0.008	0.039	0.263
no_of_weekend_nights	0.1116	0.020	5.645	0.000	0.073	0.150
no_of_week_nights	0.0444	0.012	3.608	0.000	0.020	0.068
required_car_parking_space	-1.5778	0.138	-11.453	0.000	-1.848	-1.308
lead_time	0.0155	0.000	58.788	0.000	0.015	0.016
arrival_year	0.4710	0.060	7.901	0.000	0.354	0.588
arrival_month	-0.0406	0.006	-6.290	0.000	-0.053	-0.028
arrival_date	0.0005	0.002	0.259	0.796	-0.003	0.004
repeated_guest	-2.2488	0.619	-3.632	0.000	-3.462	-1.035
no_of_previous_cancellations	0.2573	0.086	3.002	0.003	0.089	0.425
no_of_previous_bookings_not_canceled	-0.1729	0.151	-1.146	0.252	-0.469	0.123
avg_price_per_room	0.0188	0.001	25.659	0.000	0.017	0.020
no_of_special_requests	-1.4652	0.030	-48.811	0.000	-1.524	-1.406
type_of_meal_plan_Meal Plan 2	0.1632	0.066	2.461	0.014	0.033	0.293
type_of_meal_plan_Not Selected	0.2819	0.053	5.325	0.000	0.178	0.386
room_type_reserved_Room_Type 2	-0.3539	0.130	-2.713	0.007	-0.610	-0.098
room_type_reserved_Room_Type 4	-0.2570	0.053	-4.842	0.000	-0.361	-0.153
room_type_reserved_Room_Type 5	-0.7263	0.208	-3.489	0.000	-1.134	-0.318
room_type_reserved_Room_Type 6	-0.9387	0.147	-6.399	0.000	-1.226	-0.651
room_type_reserved_Room_Type 7	-1.3941	0.292	-4.767	0.000	-1.967	-0.821
market_segment_type_Corporate	0.9472	0.106	8.921	0.000	0.739	1.155
market_segment_type_Online	1.7489	0.051	34.005	0.000	1.648	1.850

Table 9. Logistic regression result 4

7.1.2.5 Dropping arrival date

Dropping the column with p = 0.796

Logit Regression Results

Dep. Variable:	booking_status	No. Observations:	25392
Model:	Logit	Df Residuals:	25369
Method:	MLE	Df Model:	22
Date:	Sat, 10 Aug 2024	Pseudo R-squ.:	0.3272
Time:	16:06:36	Log-Likelihood:	-10826.
converged:	True	LL-Null:	-16091.
Covariance Type:	nonrobust	LLR p-value:	0.000

	coef	std err	z	P>|z|	[0.025	0.975]
const	-955.0165	120.286	-7.940	0.000	-1190.773	-719.260
no_of_adults	0.0902	0.037	2.412	0.016	0.017	0.163
no_of_children	0.1515	0.057	2.652	0.008	0.040	0.264
no_of_weekend_nights	0.1117	0.020	5.655	0.000	0.073	0.150
no_of_week_nights	0.0443	0.012	3.605	0.000	0.020	0.068
required_car_parking_space	-1.5780	0.138	-11.454	0.000	-1.848	-1.308
lead_time	0.0155	0.000	58.800	0.000	0.015	0.016
arrival_year	0.4710	0.060	7.901	0.000	0.354	0.588
arrival_month	-0.0407	0.006	-6.315	0.000	-0.053	-0.028
repeated_guest	-2.2501	0.619	-3.633	0.000	-3.464	-1.036
no_of_previous_cancellations	0.2571	0.086	3.000	0.003	0.089	0.425
no_of_previous_bookings_not_canceled	-0.1727	0.151	-1.145	0.252	-0.468	0.123
avg_price_per_room	0.0188	0.001	25.659	0.000	0.017	0.020
no_of_special_requests	-1.4650	0.030	-48.820	0.000	-1.524	-1.406
type_of_meal_plan_Meal Plan 2	0.1636	0.066	2.467	0.014	0.034	0.294
type_of_meal_plan_Not Selected	0.2821	0.053	5.329	0.000	0.178	0.386
room_type_reserved_Room_Type 2	-0.3533	0.130	-2.709	0.007	-0.609	-0.098
room_type_reserved_Room_Type 4	-0.2568	0.053	-4.839	0.000	-0.361	-0.153
room_type_reserved_Room_Type 5	-0.7260	0.208	-3.488	0.000	-1.134	-0.318
room_type_reserved_Room_Type 6	-0.9386	0.147	-6.398	0.000	-1.226	-0.651
room_type_reserved_Room_Type 7	-1.3937	0.292	-4.766	0.000	-1.967	-0.821
market_segment_type_Corporate	0.9485	0.106	8.944	0.000	0.741	1.156
market_segment_type_Online	1.7492	0.051	34.016	0.000	1.648	1.850

Table 10. Logistic regression result 5

7.1.2.6 Dropping number of previous bookings not cancelled

Dropping the column with p = 0.252

Logit Regression Results

Dep. Variable:	booking_status	No. Observations:	25392
Model:	Logit	Df Residuals:	25370
Method:	MLE	Df Model:	21
Date:	Sat, 10 Aug 2024	Pseudo R-squ.:	0.3271
Time:	16:07:21	Log-Likelihood:	-10828.
converged:	True	LL-Null:	-16091.
Covariance Type:	nonrobust	LLR p-value:	0.000

	coef	std err	z	P>|z|	[0.025	0.975]
const	-953.2342	120.296	-7.924	0.000	-1189.011	-717.458
no_of_adults	0.0902	0.037	2.411	0.016	0.017	0.163
no_of_children	0.1515	0.057	2.651	0.008	0.040	0.264
no_of_weekend_nights	0.1117	0.020	5.651	0.000	0.073	0.150
no_of_week_nights	0.0443	0.012	3.603	0.000	0.020	0.068
required_car_parking_space	-1.5774	0.138	-11.449	0.000	-1.847	-1.307
lead_time	0.0155	0.000	58.843	0.000	0.015	0.016
arrival_year	0.4701	0.060	7.885	0.000	0.353	0.587
arrival_month	-0.0407	0.006	-6.314	0.000	-0.053	-0.028
repeated_guest	-2.6664	0.560	-4.763	0.000	-3.764	-1.569
no_of_previous_cancellations	0.2214	0.077	2.878	0.004	0.071	0.372
avg_price_per_room	0.0189	0.001	25.670	0.000	0.017	0.020
no_of_special_requests	-1.4655	0.030	-48.839	0.000	-1.524	-1.407
type_of_meal_plan_Meal Plan 2	0.1626	0.066	2.452	0.014	0.033	0.293
type_of_meal_plan_Not Selected	0.2821	0.053	5.330	0.000	0.178	0.386
room_type_reserved_Room_Type 2	-0.3536	0.130	-2.711	0.007	-0.609	-0.098
room_type_reserved_Room_Type 4	-0.2570	0.053	-4.843	0.000	-0.361	-0.153
room_type_reserved_Room_Type 5	-0.7261	0.208	-3.489	0.000	-1.134	-0.318
room_type_reserved_Room_Type 6	-0.9392	0.147	-6.401	0.000	-1.227	-0.652
room_type_reserved_Room_Type 7	-1.3944	0.292	-4.767	0.000	-1.968	-0.821
market_segment_type_Corporate	0.9435	0.106	8.891	0.000	0.736	1.151
market_segment_type_Online	1.7492	0.051	34.017	0.000	1.648	1.850

Table 11. Logistic regression result 6

OBSERVATIONS:

- As all the p values are lesser than 0.05, we can consider features in **X_train6** as the final ones and **lg6** as the final model.

7.1.3 Coefficient interpretations

	const	no_of_adults	no_of_children	no_of_weekend_nights	no_of_week_nights	required_car_parking_space	lead_time	arrival_year	arrival_month
Odds	0.0	1.094354	1.163622	1.118157	1.045295	0.206516	1.015634	1.600170	0.960125
Change_odd%	-100.0	9.435388	16.362210	11.815692	4.529470	-79.348388	1.563381	60.017041	-3.987489

repeated_guest	no_of_previous_cancellations	avg_price_per_room	no_of_special_requests	type_of_meal_plan_Meal Plan 2	type_of_meal_plan_Not Selected
0.069504	1.247883	1.019032	0.230951	1.176535	1.325955
-93.049626	24.788293	1.903157	-76.904870	17.653522	32.595485

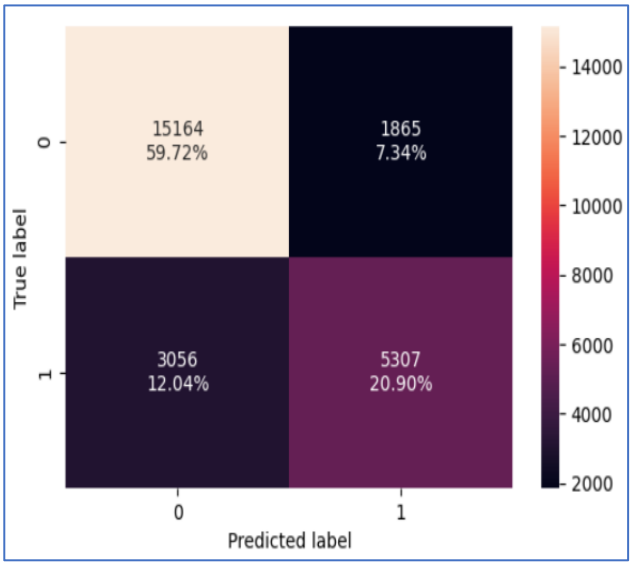
room_type_reserved_Room_Type 2	room_type_reserved_Room_Type 4	room_type_reserved_Room_Type 5	room_type_reserved_Room_Type 6	room_type_reserved_Room_Type 7
0.702174	0.773333	0.483801	0.390935	0.247982
-29.782593	-22.666659	-51.619922	-60.906525	-75.201758

market_segment_type_Online
5.749978
474.997823

OBSERVATIONS:

- No of adults: Holding all other features constant a unit change in no of adults will increase the odds of a customer cancelling the booking by 1.09 times or a 9.43% increase in odds.
- No of children: Holding all other features constant a unit change in no of children will increase the odds of a customer cancelling the booking by 1.16 times or a 16% increase in odds.
- No of weekend nights: The odds of a customer cancelling the booking on a weekend night is 1.11 times.
- No of week nights: The odds of a customer cancelling the booking on a week night is 11%.
- Required car parking space: Holding all other features constant a unit change in car parking space will decrease the odds of a customer cancelling the booking by 0.02 times or a 79% increase in odds.
- Lead time: The odds of a customer cancelling the booking on a given lead time id 1.01 times.
- No of previous cancellations: Number of previous cancelations contributes to 24% increase.
- Average price per room: Holding all other features constant a unit change in average price per room will increase the odds of a customer cancelling the booking by 1.01 times or a 1.90% increase in odds.
- Meal plan type: The meal plan 2 contributes to 17% and customers who have not selected the meal plan have 32% chances to cancel the booking.
- Room type reserved: The room type 7 decreases the chance of cancelling the booking by 75%.
- Market segment type corporate: The online segment increases the chances of cancelling by 474% and corporate by 156%.

7.1.4 Checking model performance on the training set



Training performance:

	Accuracy	Recall	Precision	F1
0	0.806199	0.634581	0.739961	0.683231

Fig 39. Confusion matrix-Train set

OBSERVATIONS:

- The false negatives constituted to around 12.04%.
- The model has a good accuracy of 0.80 and relatively low recall of 0.63.

7.1.4.1 ROC-AUC

The ROC-AUC curve is a performance measurement tool for binary classification models. It helps to evaluate how well a model distinguishes between classes, with a focus on the trade-off between the True Positive Rate (TPR) and the False Positive Rate (FPR) across different threshold values.

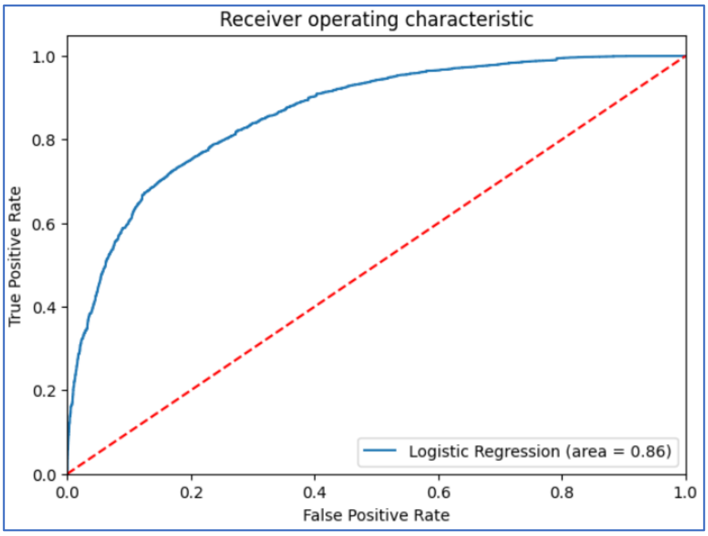


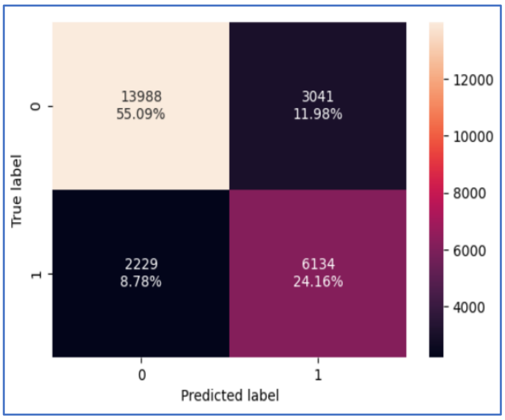
Fig 40. ROC-AUC-Train set

OBSERVATIONS:

- The model is performing well on training set.

7.1.4.2 Optimal threshold using ROC-AUC curve

The optimum threshold using the ROC-AUV curve is **0.3731**



Training performance:

	Accuracy	Recall	Precision	F1
0	0.792454	0.733469	0.668556	0.69951

Fig 41. Confusion matrix-Train set Optimum threshold

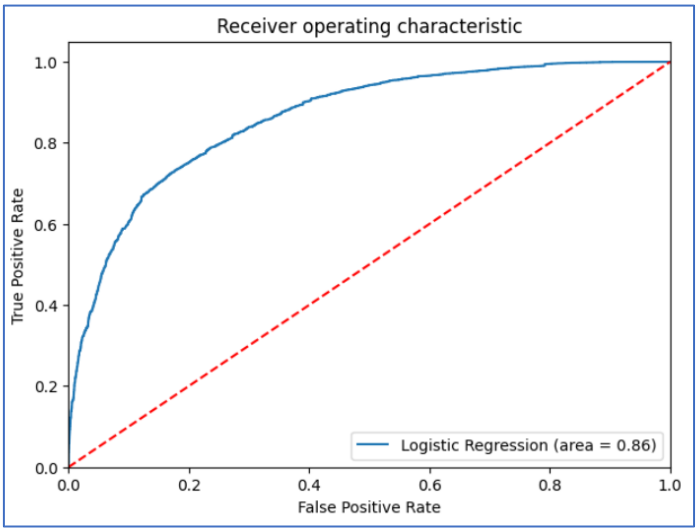


Fig 42. ROC-AUC-Train set Optimum threshold

OBSERVATIONS:

- The false negatives have reduced from 12% to 8.78%.
- Recall of model and F1 have increased to 0.73 and 0.69, but the other metrics have reduced.
- The model is still giving a good performance.

7.1.4.3 Precision-Recall Curve

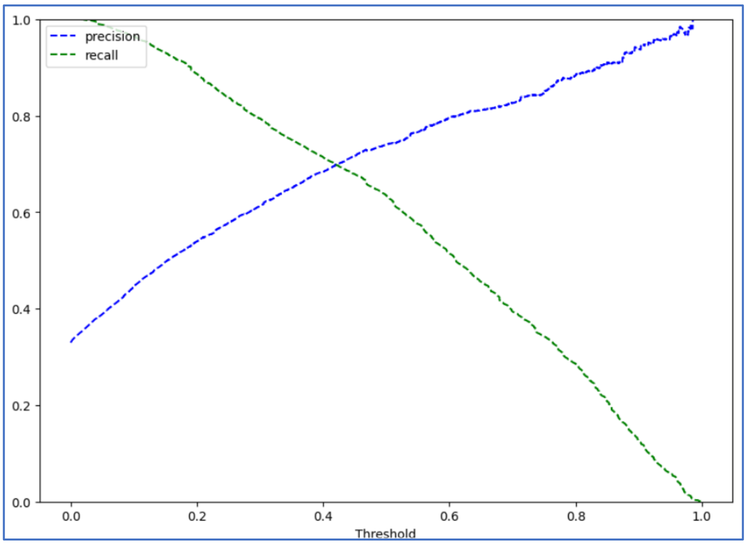
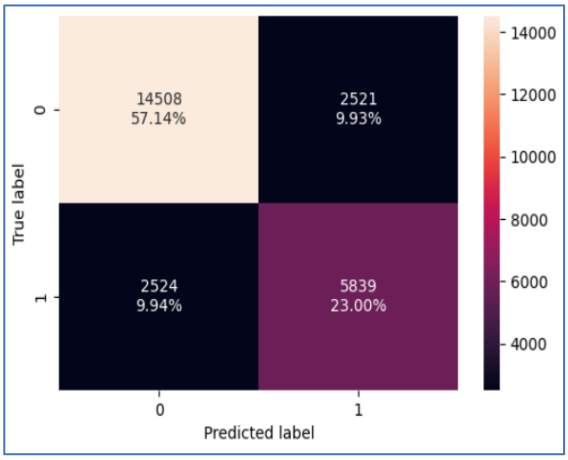


Fig 43. Precision-Recall curve

OBSERVATIONS:

- At the threshold of 0.42, we get balanced recall and precision.



Training performance:

	Accuracy	Recall	Precision	F1
0	0.801315	0.698194	0.698445	0.69832

Fig 44. Confusion matrix-Train set Threshold

OBSERVATIONS:

- The recall has reduced from 0.73 to 0.69.
- There's not much improvement in the model performance.

7.1.4.4 Training performance comparison

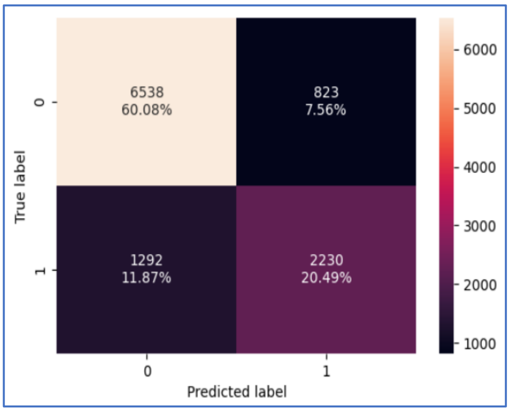
Training performance comparison:

	Logistic Regression statsmodel	Logistic Regression-0.37 Threshold	Logistic Regression-0.42 Threshold
Accuracy	0.806199	0.792454	0.801315
Recall	0.634581	0.733469	0.698194
Precision	0.739961	0.668556	0.698445
F1	0.683231	0.699510	0.698320

OBSERVATIONS:

- The recall is higher (0.73), with a threshold of 0.37.
- There is no much difference is the accuracy.

7.1.5 Checking model performance on the test set



Test performance:

	Accuracy	Recall	Precision	F1
0	0.80566	0.633163	0.730429	0.678327

Fig 45. Confusion matrix-Test set

OBSERVATIONS:

- The model has a good accuracy of 0.80 and precision of 0.73 and relatively low recall of 0.63.
- The false negatives account to about 11.87%.

7.1.5.1 ROC-AUC

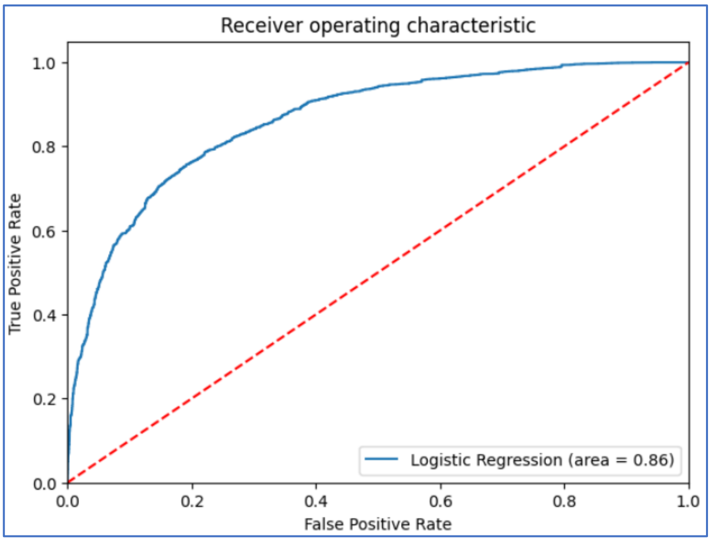
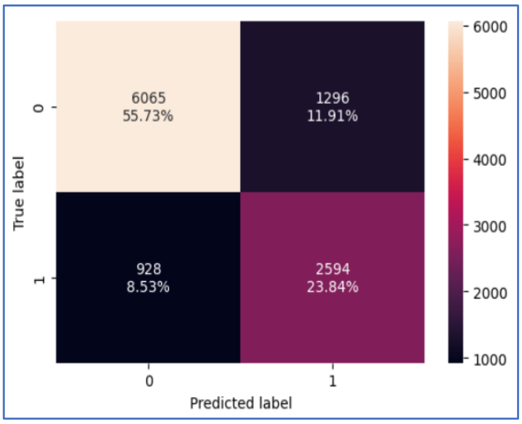


Fig 46. ROC-AUC-Test set

OBSERVATIONS:

- The model is performing well on test set.

7.1.5.2 Optimal threshold using ROC-AUC curve



Test performance:

	Accuracy	Recall	Precision	F1
0	0.795645	0.736513	0.666838	0.699946

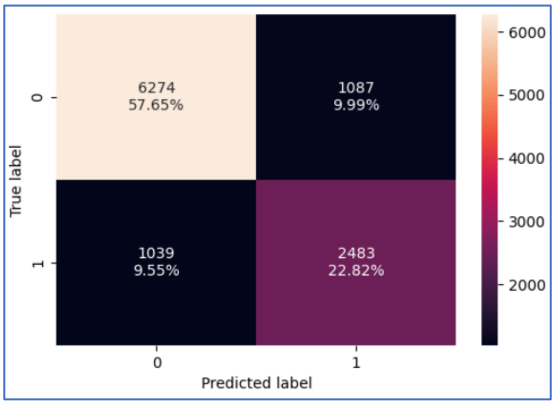
Fig 46. Confusion matrix-Test set Optimum threshold

OBSERVATIONS:

- The recall has improved significantly, increasing from 0.63 to 0.73.

7.1.5.3 Precision-Recall Curve

At the threshold of 0.42, we get the below matrix.



Test performance:

	Accuracy	Recall	Precision	F1
0	0.804649	0.704997	0.695518	0.700226

Fig 47. Confusion matrix-Test set Threshold

OBSERVATIONS:

- The recall has dropped to 0.70 from 0.73.
- There's not much improvement in the model performance.

7.1.5.4 Test performance comparison

Test set performance comparison:

	Logistic Regression statsmodel	Logistic Regression-0.37 Threshold	Logistic Regression-0.42 Threshold
Accuracy	0.805660	0.795645	0.804649
Recall	0.633163	0.736513	0.704997
Precision	0.730429	0.666838	0.695518
F1	0.678327	0.699946	0.700226

OBSERVATIONS:

- The recall is higher (0.73), with a threshold of 0.37.
- The accuracy is almost consistent among the models.
- There is a slight decrease in precision across the models, while the F1 score has been observed to increase.

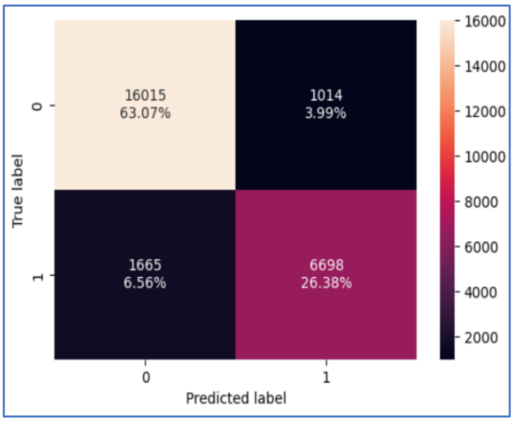
7.1.6 Conclusions

- All the models are giving a generalized performance on training and test set.
- The highest recall is 0.73 on both training and test set.
- Using the model with 0.37 threshold the model will give a high recall but low precision scores. This model will help the INN Hotels group identify potential reasons for cancellations but the cancellations will be high.
- Using the model with 0.42 threshold the model will give a balance recall and precision score. This model will help the INN Hotels group identify potential reasons for cancellations and manage the cancellations.

8. K- Nearest Neighbor (KNN) Classifier (sklearn)

In order to optimize our model, it's essential to experiment with different values of k to find the most suitable fit for our data. We can commence this process by setting k equal to 3 and gradually exploring other values to assess their impact on the model's performance.

8.1 Checking model performance on Training set



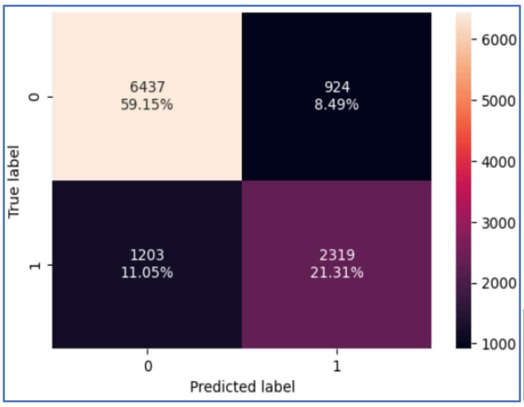
	Accuracy	Recall	Precision	F1
0	0.894494	0.800909	0.868517	0.833344

Fig 48. KNN-Confusion matrix-Train set

OBSERVATIONS:

- The percentage of false negatives is quite low, at approximately 6.56%.
- The recall and precision are high, about 0.80 and 0.86 respectively.
- The model appears to fit the data well.

8.2 Checking model performance on Test set



	Accuracy	Recall	Precision	F1
0	0.804558	0.658433	0.715079	0.685588

Fig 49. KNN-Confusion matrix-Test set

OBSERVATIONS:

- The percentage of false negatives is higher than the training set.
- The precision is high, at approximately 0.71, while the recall is lower compared to the training set.
- The model appears to fit the data well.

8.3 K with different values

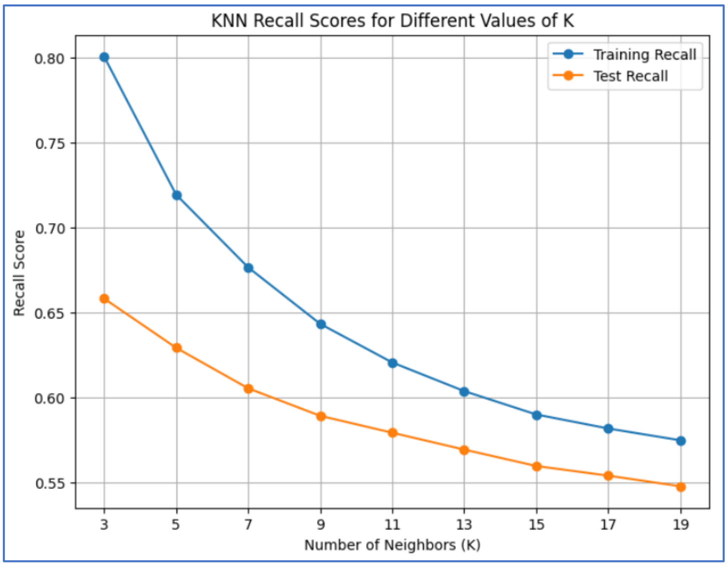


Fig 50. K with different values

OBSERVATIONS:

- Among various recall scores evaluated, a recall score of 3 is considered the best. This implies that a recall score of 3 is either a specific value that indicates optimal performance.

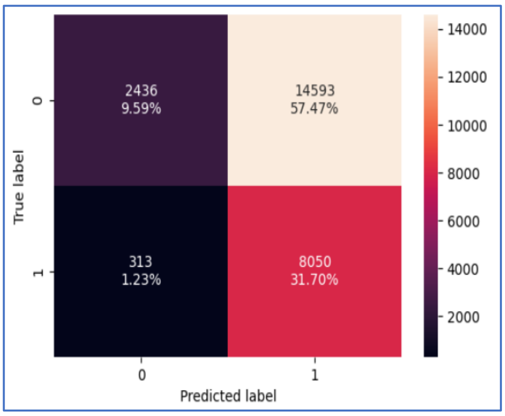
8.4 Conclusions

- The KNN classifier gives a recall of 0.80 for training set and 0.65 for test set which means that the model correctly identifies 80% and 65% of all actual positive instances that the booking will be cancelled.
- The accuracy of the both the training set and the test set is seen to be high, about 0.89 and 0.80 respectively.
- The model will help the INN Hotels group identify potential reasons for cancellations and manage the cancellations.

9. Naive-Bayes Classifier (sklearn)

The Naive Bayes classifier is a simple yet effective probabilistic classifier based on Bayes' theorem with an assumption of independence between features.

9.1 Checking model performance on Training set



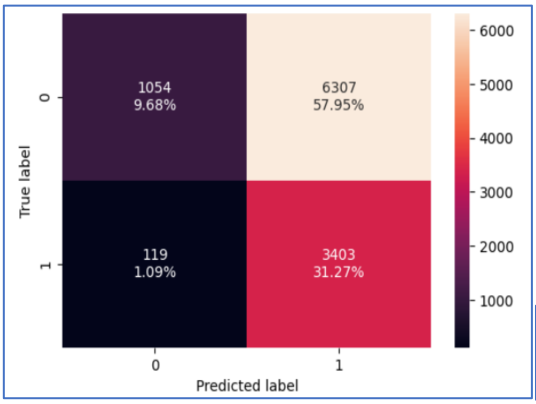
	Accuracy	Recall	Precision	F1
0	0.412965	0.962573	0.355518	0.519254

Fig 51. Naive-Bayes-Confusion matrix-Train set

OBSERVATIONS:

- The false negatives are relatively low, about 1.23%.
- The accuracy is quite low, approximately 0.41.
- The recall is relatively high at 0.96, but the precision is quite low, around 0.35.
- There is high rate of false positives which can lead to numerous false alarms and may reduce the reliability of the model's positive predictions.

9.2 Checking model performance on Test set



	Accuracy	Recall	Precision	F1
0	0.409538	0.966212	0.350463	0.514359

Fig 52. Naive-Bayes-Confusion matrix-Test set

OBSERVATIONS:

- The false negatives are relatively low, about 1.29%.
- The accuracy is quite low, approximately 0.40.
- The training and test datasets perform similarly.

9.3 Comparison of Models

Training performance comparison:

	K Nearest Neighbor k=3	Naive Bayes
Accuracy	0.894494	0.412965
Recall	0.800909	0.962573
Precision	0.868517	0.355518
F1	0.833344	0.519254

Test set performance comparison:

	K Nearest Neighbor k=3	Naive Bayes
Accuracy	0.804558	0.409538
Recall	0.658433	0.966212
Precision	0.715079	0.350463
F1	0.685588	0.514359

OBSERVATIONS:

- The KNN model performs well on the training set which gives a better recall and precision of 0.80 and 0.86 respectively.
- The Naive Bayes model gives a very good recall score (0.96), which ensures that most of the actual positive cases are identified. It has a low precision (0.35), where many of the predicted positive cases are incorrect which can lead to a high number of false positives.
- The training and test scores on the Naive Bayes model are nearly identical.

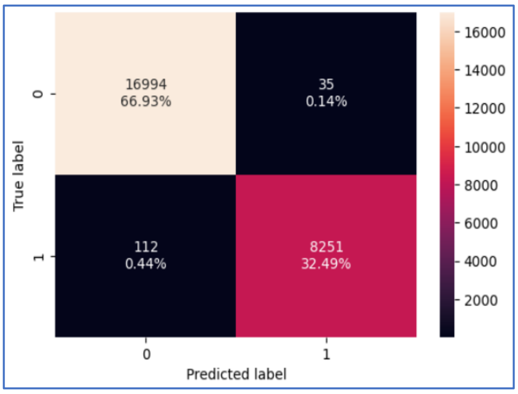
9.4 Conclusion

- Among the models evaluated, KNN is able to maintain good recall score and precision scores of about 0.65 and 0.71 respectively.
- The Naive Bayes model has very good recall but drops in precision nearly 3 times, which increases the false positives to a greater extent.
- The F1 score is comparatively higher for the KNN model.
- Considering the above, the INN Hotels group can should integrate the KNN model with k=3 into the analysis for decreasing the last minute cancellations of bookings.

10. Decision Tree Classifier (sklearn)

A Decision Tree Classifier is a popular machine learning algorithm used for classification tasks. It works by splitting the data into subsets based on feature values, aiming to create a tree-like model of decisions.

10.1 Checking model performance on Training set



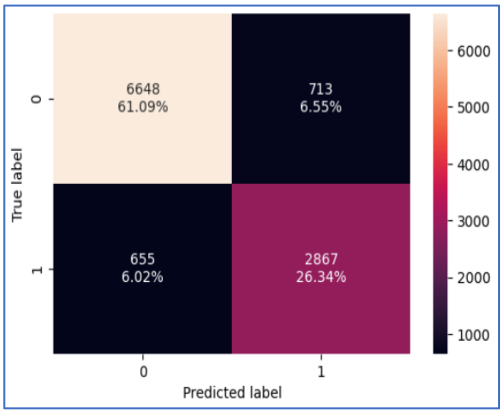
	Accuracy	Recall	Precision	F1
0	0.994211	0.986608	0.995776	0.991171

Fig 53. Decision Tree-Confusion matrix-Train set

OBSERVATIONS:

- The false negatives contribute very less in the data set, which is about 0.44%.
- The accuracy, recall, and precision are all exceptionally high, approximately 0.99, 0.98, and 0.99, respectively.
- The model fits well on the training set.

10.2 Checking model performance on Test set



	Accuracy	Recall	Precision	F1
0	0.874299	0.814026	0.800838	0.807378

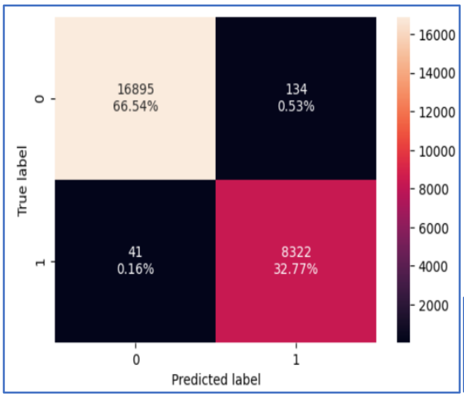
Fig 54. Decision Tree-Confusion matrix-Test set

OBSERVATIONS:

- The false negatives contribute less in the data set, which is about 6.02%.
- The accuracy, recall, and precision are all commendable, around 0.87, 0.81, and 0.80, respectively.
- The F1 score is good which is close to 1.
- The model fits well on the test set.

10.3 Decision Tree (with class_weights)

10.3.1 Checking model performance on Training set



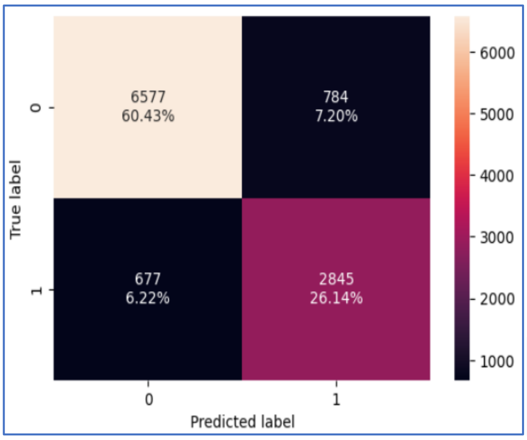
	Accuracy	Recall	Precision	F1
0	0.993108	0.995097	0.984153	0.989595

Fig 55. Decision Tree (with weights)-Confusion matrix-Train set

OBSERVATIONS:

- The false negatives contribute only about 0.16%.
- The accuracy, recall, and precision are good.
- The F1 score is good which is close to 1.
- The model fits well on the train set.

10.3.2 Checking model performance on Test set



	Accuracy	Recall	Precision	F1
0	0.865754	0.80778	0.783963	0.795693

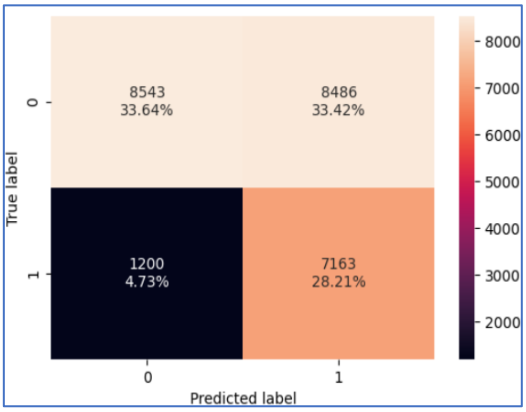
Fig 56. Decision Tree (with weights)-Confusion matrix-Test set

OBSERVATIONS:

- The accuracy, recall, and precision are good.
- The model fits well on the test set.

10.4 Decision Tree (Pre-pruning)

10.4.1 Checking model performance on Training set



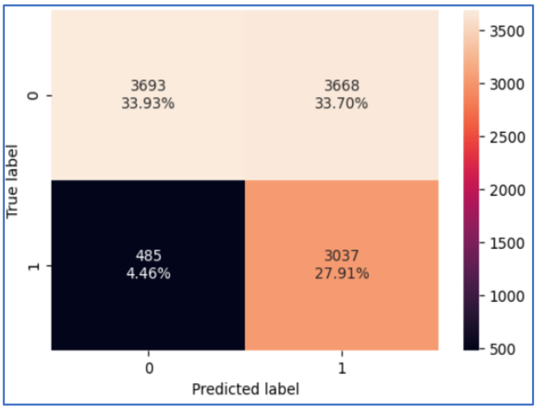
	Accuracy	Recall	Precision	F1
0	0.618541	0.856511	0.457729	0.596618

Fig 57. Decision Tree (Pre-pruning)-Confusion matrix-Train set

OBSERVATIONS:

- The accuracy, recall, and precision are found to decrease with pre-pruning.
- The recall drops from 0.98 to 0.85.
- The accuracy drops from 0.90 to 0.61.
- The model's performance is observed to decline.

10.4.2 Checking model performance on Test set



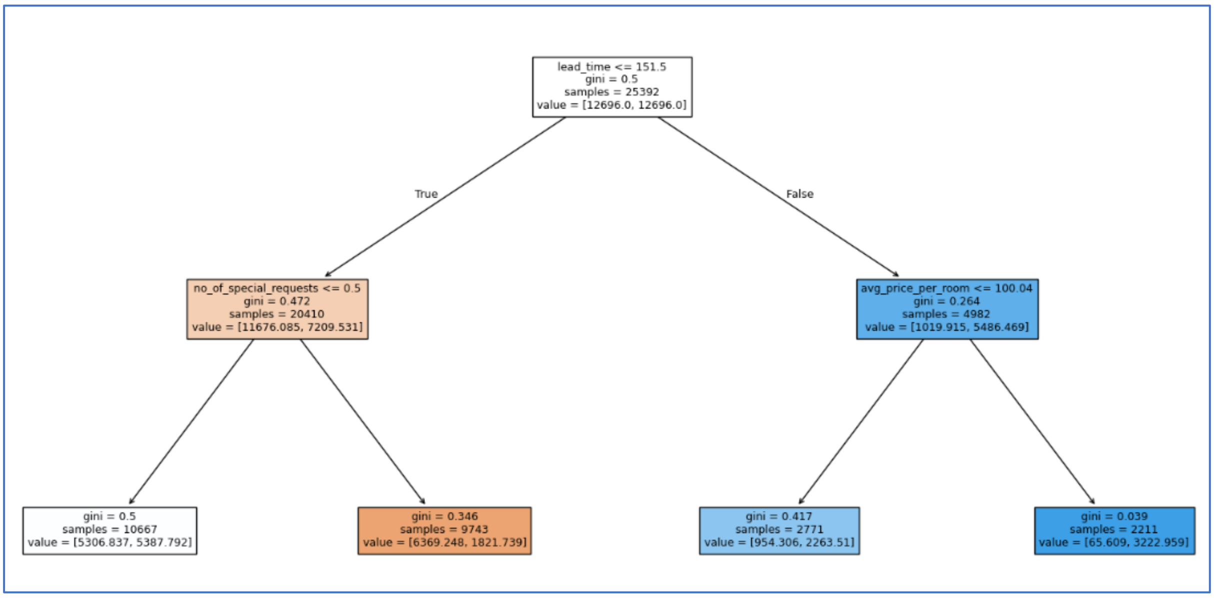
	Accuracy	Recall	Precision	F1
0	0.618396	0.862294	0.452946	0.593918

Fig 58. Decision Tree (Pre-pruning)-Confusion matrix-Test set

OBSERVATIONS:

- The accuracy, recall, and precision are found to decrease with pre-pruning.
- The recall drops from 0.81 to 0.61.
- The accuracy drops to 0.61.
- The model's performance is observed to decline.

10.4.3 Decision Tree




```
|--- lead_time <= 151.50
|   |--- no_of_special_requests <= 0.50
|   |   |--- weights: [5306.84, 5387.79] class: 1
|   |--- no_of_special_requests >  0.50
|   |   |--- weights: [6369.25, 1821.74] class: 0
|--- lead_time >  151.50
|   |--- avg_price_per_room <= 100.04
|   |   |--- weights: [954.31, 2263.51] class: 1
|   |--- avg_price_per_room >  100.04
|   |   |--- weights: [65.61, 3222.96] class: 1
```

Fig 59. Decision Tree (Pre-pruning)

OBSERVATIONS:

- If the lead time is more than 151.5 days and the average price of the room is above 100 euros, then the booking is more likely to be cancelled.

10.4.4 Important Features

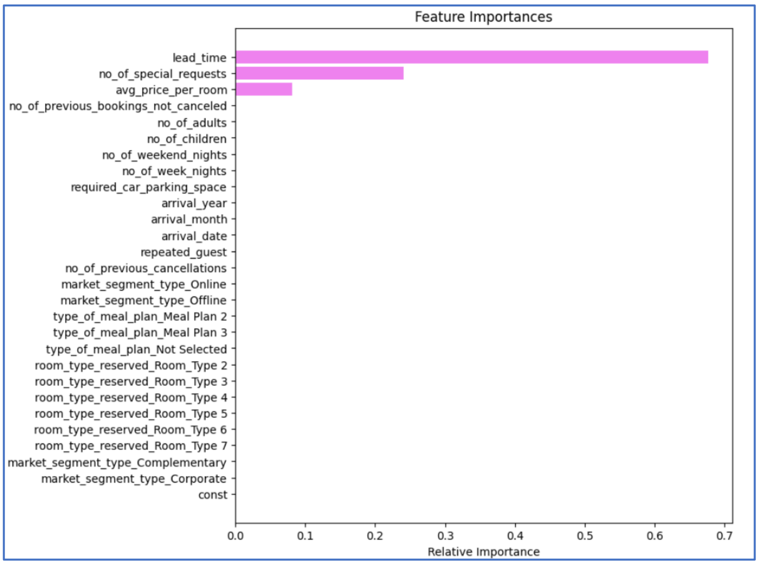


Fig 60. Important features (Pre-pruning)

OBSERVATIONS:

- In the pre tuned decision tree, lead time, number of special requests and average price per room are the most important features.

10.5 Decision Tree (Post pruning)

In Decision Tree Classifier, the pruning technique is parameterized by the cost complexity parameter, ccp_alpha. Greater values of ccp_alpha increase the number of nodes pruned. Here only the effect of ccp_alpha on regularizing the trees and how to choose a ccp_alpha based on validation scores is considered.

10.5.1 Total impurity vs effective alpha for training set

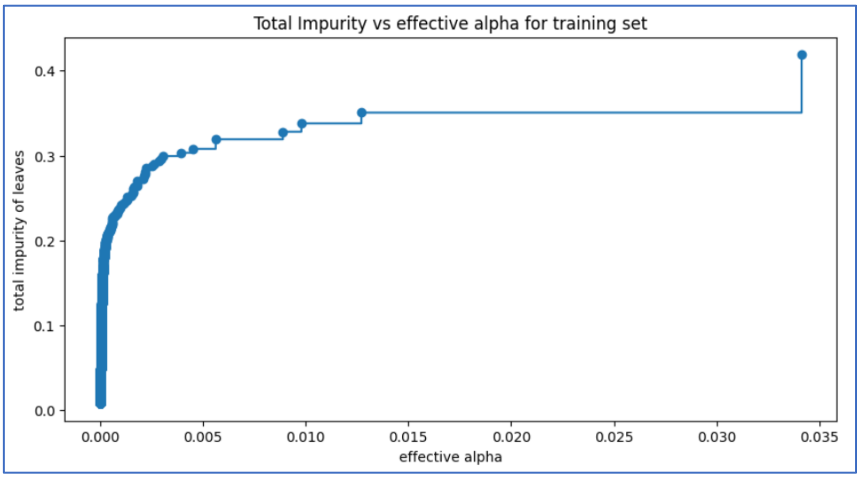


Fig 61. Total impurity vs effective alpha-Train set

OBSERVATIONS:

- The total impurity of is minimized at ccp_alpha of 0.0811.

10.5.2 Number of nodes/Depth vs alpha

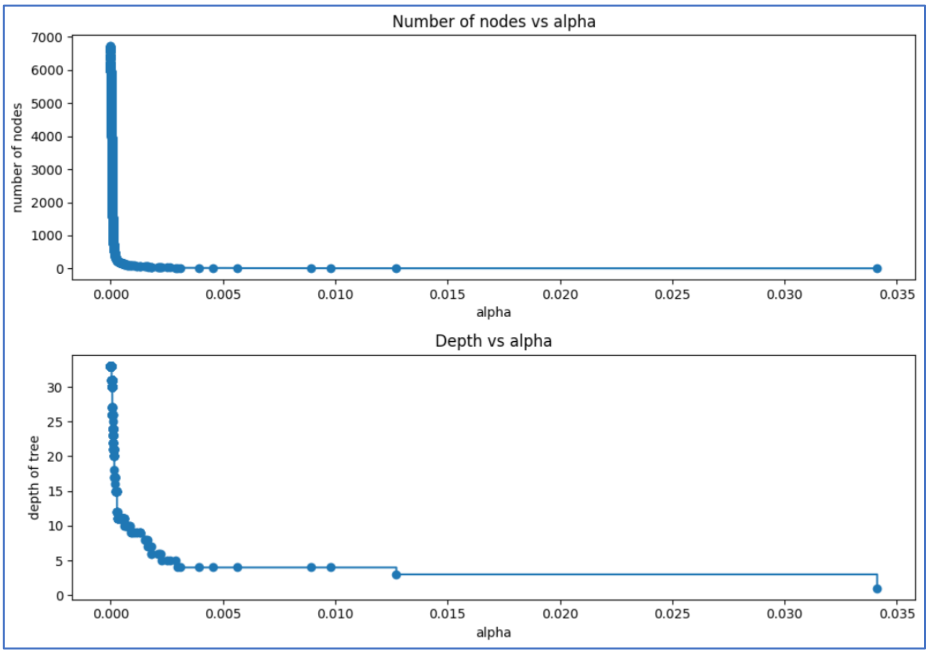


Fig 62. Number of nodes/Depth vs alpha-Train set

OBSERVATIONS:

- As alpha increases, the number of nodes generally decreases. The higher alpha values result in more pruning, removing branches and nodes to simplify the tree.
- As alpha increases, the depth of the tree generally decreases. The higher alpha values result in more pruning, which removes branches and reduces the tree's depth.

10.5.3 Recall vs alpha for training and test set

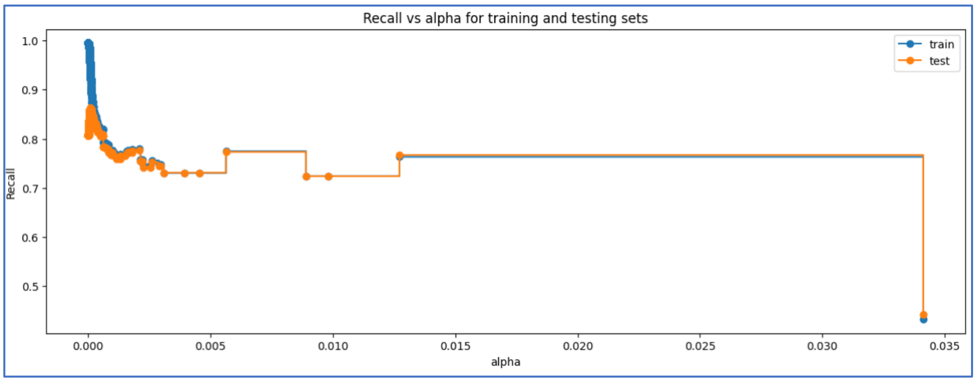
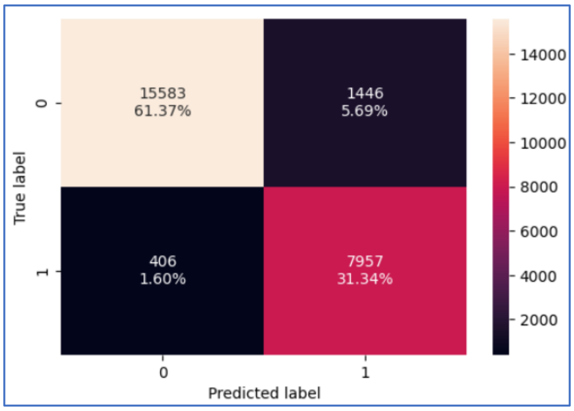


Fig 63. Recall vs alpha-Train and Test set

OBSERVATIONS:

- The recall for train and test data seem to fit well and perform good at the ccp_alpha of 0.08.

10.5.4 Checking model performance on training set



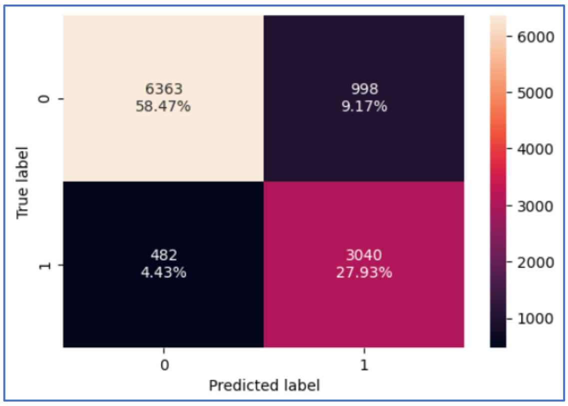
	Accuracy	Recall	Precision	F1
0	0.927064	0.951453	0.846219	0.895756

Fig 64. Decision Tree (Post-pruning)-Confusion matrix-Train set

OBSERVATIONS:

- The recall, precision and accuracy are high with values of 0.95, 0.84 and 0.92 respectively.
- The model seems to fit well.

10.5.5 Checking model performance on test set



	Accuracy	Recall	Precision	F1
0	0.864008	0.863146	0.752848	0.804233

Fig 65. Decision Tree (Post-pruning)-Confusion matrix-Test set

OBSERVATIONS:

- The recall, precision and accuracy are good with the test data.
- The model seems to fit well.

10.5.6 Decision tree (Post-pruning)

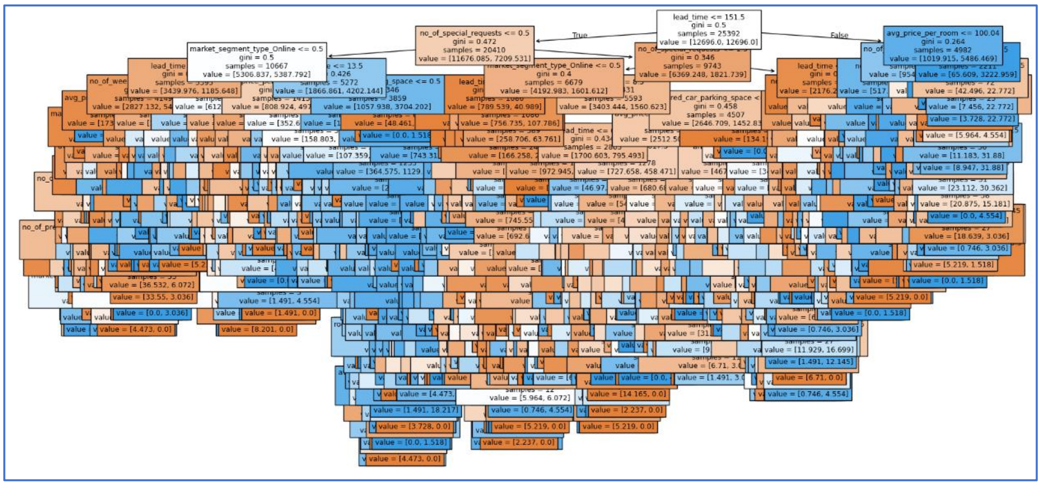


Fig 66. Decision Tree (Post-pruning)

OBSERVATIONS:

- The decision tree is very complex having a high depth and a large number of nodes. This complexity can lead to several issues, including overfitting and interpretability challenges.

10.5.7 Important features

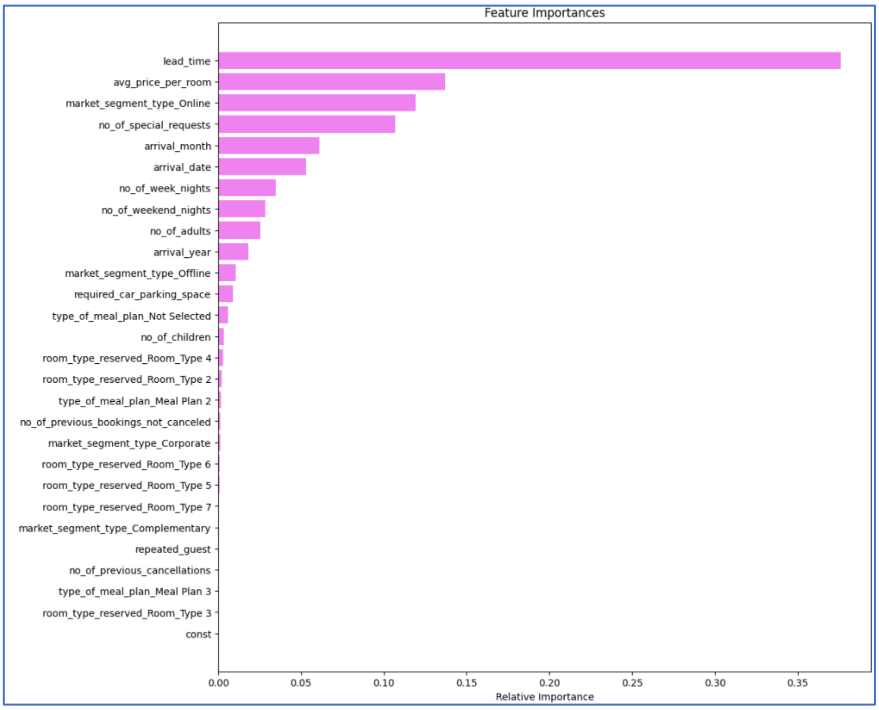


Fig 67. Important features (Post-pruning)

OBSERVATIONS

- In the post tuned decision tree, lead time, average price per room, market segment online, number of special requests, arrival month and arrival date are the most important features.

10.6 Decision tree model comparison for Training set

Training performance comparison:

	Decision Tree without class_weight	Decision Tree with class_weight	Decision Tree (Pre-Pruning)	Decision Tree (Post-Pruning)
Accuracy	0.994211	0.993108	0.618541	0.927064
Recall	0.986608	0.995097	0.856511	0.951453
Precision	0.995776	0.984153	0.457729	0.846219
F1	0.991171	0.989595	0.596618	0.895756

OBSERVATIONS:

- The decision tree without class weights demonstrates strong performance, with high accuracy, recall, and precision, outperforming the decision tree with class weights.
- However, both pre-pruning and post-pruning significantly reduce the model's recall and precision.

10.7 Decision tree model comparison for Test set

Test set performance comparison:

	Decision Tree without class_weight	Decision Tree with class_weight	Decision Tree (Pre-Pruning)	Decision Tree (Post-Pruning)
Accuracy	0.874299	0.865754	0.618396	0.864008
Recall	0.814026	0.807780	0.862294	0.863146
Precision	0.800838	0.783963	0.452946	0.752848
F1	0.807378	0.795693	0.593918	0.804233

OBSERVATIONS:

- The decision tree without class weights demonstrates strong performance, with high accuracy, recall, and precision, outperforming the decision tree with class weights.
- The decision tree with post-pruning has a good recall, precision and accuracy.
- However, pre-pruning significantly reduces the model's recall and precision.

10.8 Conclusion

- The INN Hotels group can consider Decision Tree model without weights or Decision Tree model (Post-pruning) as they give good recall, precision and accuracy.

11. Model Performance Comparison and Final Model Selection

11.1 Logistic Regression

Training performance comparison:

	Logistic Regression statsmodel	Logistic Regression-0.37 Threshold	Logistic Regression-0.42 Threshold
Accuracy	0.806199	0.792454	0.801315
Recall	0.634581	0.733469	0.698194
Precision	0.739961	0.668556	0.698445
F1	0.683231	0.699510	0.698320

Test set performance comparison:

	Logistic Regression statsmodel	Logistic Regression-0.37 Threshold	Logistic Regression-0.42 Threshold
Accuracy	0.805660	0.795645	0.804649
Recall	0.633163	0.736513	0.704997
Precision	0.730429	0.666838	0.695518
F1	0.678327	0.699946	0.700226

OBSERVATIONS:

- Logistic regression model has a high recall, approximately 0.73 and a precision of 0.66 at a threshold of 0.37.
- F1 score is 0.699.

11.2 KNN and NAIVE BAYES

Training performance comparison:

	K Nearest Neighbor k=3	Naive Bayes
Accuracy	0.894494	0.412965
Recall	0.800909	0.962573
Precision	0.868517	0.355518
F1	0.833344	0.519254

Test set performance comparison:

	K Nearest Neighbor k=3	Naive Bayes
Accuracy	0.804558	0.409538
Recall	0.658433	0.966212
Precision	0.715079	0.350463
F1	0.685588	0.514359

OBSERVATIONS:

- KNN model has a recall of 0.65, a precision of 0.71 and a F1 score of 0.68.
- Naive Bayes model has a recall of 0.96, but a low precision of 0.35 and a F1 score of 0.51.

11.3 Decision Tree

Training performance comparison:

	Decision Tree without class_weight	Decision Tree with class_weight	Decision Tree (Pre-Pruning)	Decision Tree (Post-Pruning)
Accuracy	0.994211	0.993108	0.618541	0.927064
Recall	0.986608	0.995097	0.856511	0.951453
Precision	0.995776	0.984153	0.457729	0.846219
F1	0.991171	0.989595	0.596618	0.895756

Test set performance comparison:

	Decision Tree without class_weight	Decision Tree with class_weight	Decision Tree (Pre-Pruning)	Decision Tree (Post-Pruning)
Accuracy	0.874299	0.865754	0.618396	0.864008
Recall	0.814026	0.807780	0.862294	0.863146
Precision	0.800838	0.783963	0.452946	0.752848
F1	0.807378	0.795693	0.593918	0.804233

OBSERVATIONS:

- Decision tree model without class weight has a recall of 0.81, a precision of 0.80 and a F1 score of 0.80.

12. Insights

- A high recall value for cancellations of hotel room bookings indicates that the model is very effective at identifying bookings that are likely to be cancelled. The model successfully detects most of the cancellations, minimizing the number of false negatives.
- A high precision value for cancellations of hotel room bookings means that when the model predicts a booking will be cancelled, it is highly accurate. A high percentage of the predicted cancellations are indeed true cancellations, minimizing the number of false positives.
- For a model to perform well, it is necessary to have low false negatives and false positives.
- Given the criteria above, the Decision Tree model without class weights and the Post-pruned Decision Tree model both perform well on the training and test sets, exhibiting high recall, precision, accuracy, and an F1 score close to 1.
- The next preferred model would be the logistic regression model.
- The INN Hotels group could use the Decision Tree model to identify and analyse the key factors contributing to booking cancellations.

greatlearning
Learning for Life

13. Actionable Insights & Recommendations

- From the logistic regression model, it has been identified market segment type Online and Corporate is a significant predictor of a booking being cancelled. Offering a lower rate for non-refundable bookings can encourage customers to commit to their reservations.
- The cancellations happen at a higher rate for the customers who book rooms for weekend nights, as weekend bookings might be more susceptible to changes in personal or work schedules, leading to more cancellations. Enhancing the appeal of weekend bookings by offering additional services, such as early check-in, late check-out, or special weekend packages, might encourage customers to stick to their plans.
- The Decision Tree model highlights that lead time has a significant impact on cancellations. Providing discounts or special rates for customers who book well in advance. This can encourage early commitment and reduce the likelihood of cancellations.
- Bookings including children are more prone to cancellations. Families may prefer accommodations with flexible booking policies to adjust their plans without incurring penalties. If flexibility is lacking, they might cancel to avoid potential issues.
- Customers choosing meal plan 2 cancel their booking often compared to people opting the other preferences. Hence offering a range of cuisines and dishes to cater to different cultural preferences and dietary restrictions by including vegetarian, vegan, gluten-free, and allergy-friendly options.
- Customers requiring a car parking space cancel at a lower rate. Offering parking as part of the booking package or including it in the room rate can help manage expectations.
- The average price per room also contributes to cancellations. Allowing guests to lock in a rate when booking early, which can reduce cancellations due to price changes.
- Provide exclusive benefits for guests who commit to their bookings, such as room upgrades or additional services.
- Considering customer feedback would be valuable for conducting further analysis to reduce cancellations.